

SOK3022

Topic 2: Time Series Analysis – Modelling Volatility

Application to crude oil price and stock prices of oil/clean energy companies

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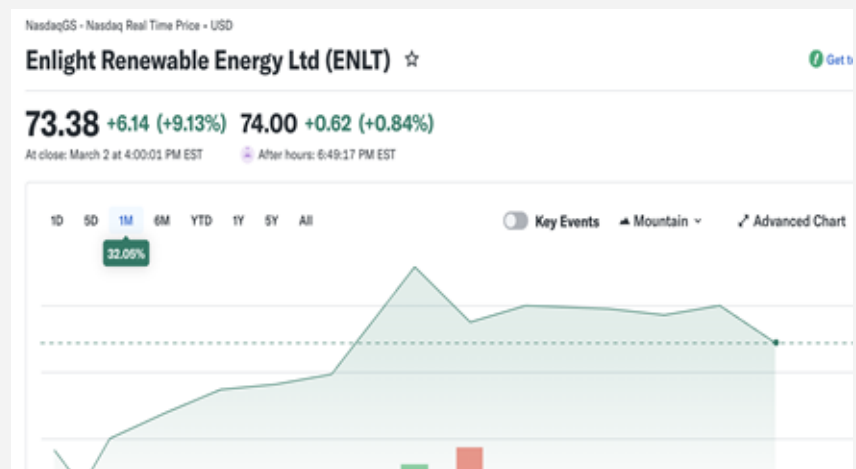
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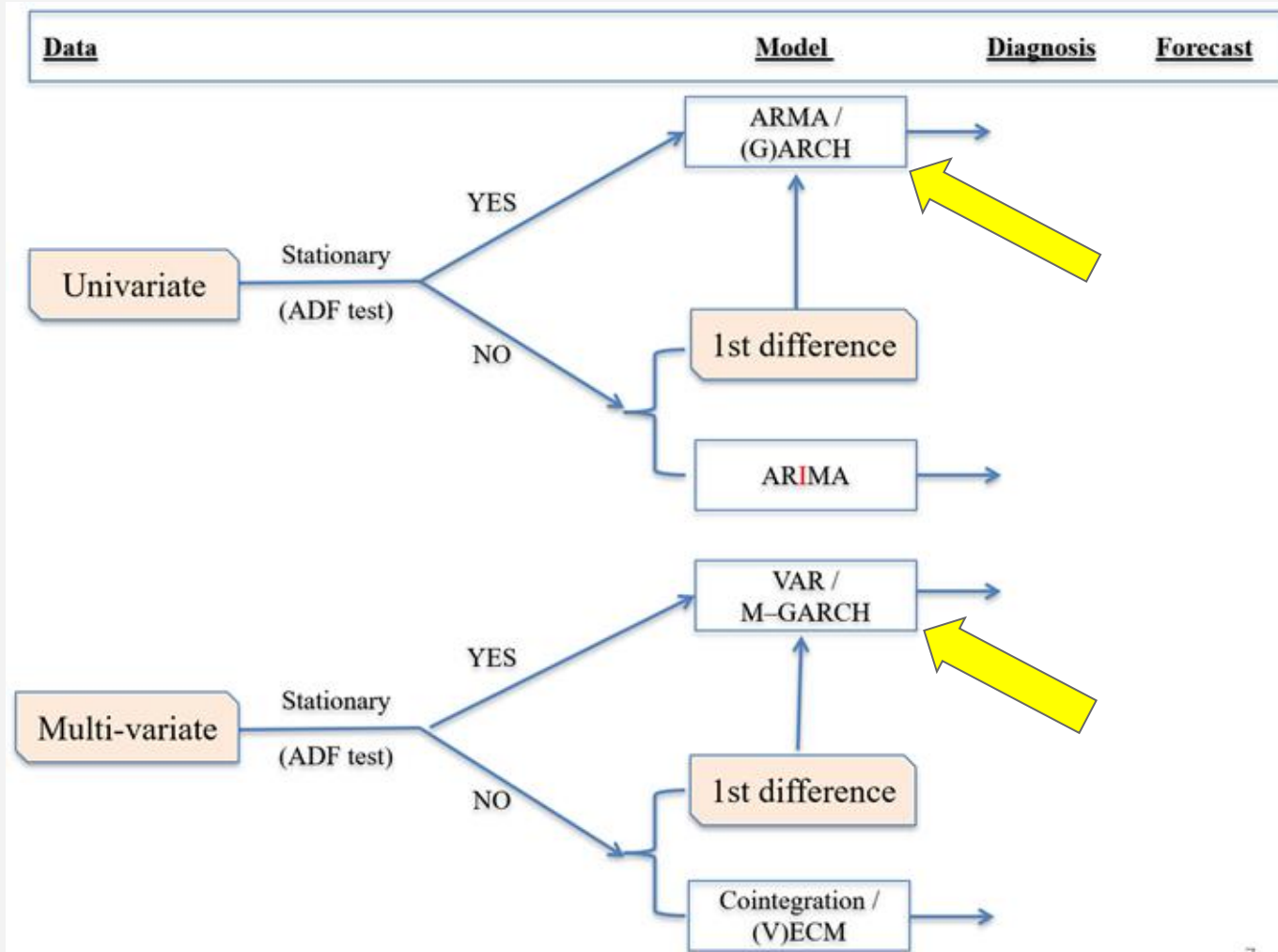
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Quiz / Discussion

- Do you think crude oil price and stock shares of renewable energy companies positively or negatively correlated?
- Do you think (or guess) stock returns are stationary or not?
- What does heteroskedasticity mean?
- Is DCC-GARCH model about volatility spillover?





Time series models

- Univariate time series: Forecast mean
 - AR (Auto-Regressive)
 - MR (Moving Average)
 - ARMA, ARIMA
- Univariate time series: Forecast volatility
 - ARCH
 - GARCH
 - T-GARCH/E-GARCH
- Multivariate time series
 - M-GARCH
 - Volatility modeling: DCC-GARCH

Lecture 1: Intro

Volatility

Return volatility is, of course, central to financial economics. Indeed, as noted by Campbell, Lo and MacKinlay (1997):

“... what distinguishes financial economics is the central role that uncertainty plays in both financial theory and its empirical implementation... Indeed in the absence of uncertainty, the problems of financial economics reduce to exercises in basic microeconomics” (p. 3).

Andersen, T. G., Bollerslev, T., Christoffersen, P. F., & Diebold, F. X. (2006). Volatility and correlation forecasting. Handbook of economic forecasting, 1, 777-878.

Volatility as a shorthand for conditional variance.

“ Modeling and forecasting volatility lies at the heart of modern finance because good estimates of correlation and volatility are needed for derivative pricing, portfolio optimization, risk management, and hedging.”
(Sadorsky, 2012, p. 248)

Volatility

- Volatility of the stock market indices/shares is a statistical measure of **return** dispersion.
- Represents the range of value changes.
- Commonly measured using standard deviation or variance of returns: **unconditional** or **conditional**.
- Indicates the degree of fluctuation in market performance.

(Zhang et al., 2018; Lim et al., 2023)

$$\ln\left(\frac{P_t}{P_{t-1}}\right) \times 100$$

Volatility: Modeling and applications

- Engle, R. F. (1982). *Autoregressive Conditional Heteroskedasticity with Estimates of the Variance of United Kingdom Inflation*. **Econometrica**, **50**(4), 987–1007.
- Bollerslev, T. (1986). *Generalized Autoregressive Conditional Heteroskedasticity*. **Journal of Econometrics**, **31**(3), 307–327.

Example

Table 1. Key Studies on Volatility Spillover in India over the Years.

Author and Year	Objective	Methodology
Kaur (2004)	Nature of stock market volatility in the Indian stock market	ARCH and GARCH
Karmakar (2005)	Conditional volatility in the Indian stock market	GARCH
Karmakar (2007)	Asymmetric volatility in the Indian stock market	EGARCH
Mishra et al. (2007)	Volatility spillovers between the Indian stock and foreign exchange markets	ARCH and E-GARCH
Mukherjee and Mishra (2010)	Volatility spillovers between India and other Asian Stock markets	GARCH model
Yilmaz (2010)	Volatility spillover in East Asian stock markets	VAR model
Jebzan and Iqbal (2016)	Volatility spillover effects between the stock market and foreign exchange market in selected Asian countries; Pakistan, India, Sri Lanka, China, Hong Kong and Japan	EGARCH
Mitra (2017)	Volatility spillovers between the Indian stock and foreign exchange markets	GARCH model
Nandy and Chattopadhyay (2019)	Interdependence between Indian stock and other domestic markets	(DCC-MV-TARCH) (1, 1) model
Sahoo et al. (2019)	Returns spillovers between currency and bond markets	BEKK-GARCH
Uludag and Khurshid (2019)	Volatility spillover from the Chinese stock market to E7 stock market	GARCH
Yadav et al. (2020)	Volatility spillover from equity markets to commodities market in India	Granger causality and dynamic conditional correlation
Dey and Sampath (2020)	Volatility spillovers in three sectors of the Indian stock market	VAR model
Ngene (2021)	Time-varying and directional connectedness of nine equity sectors through intra- and inter-sector volatility spillover in the US	Quantile regression
Oliyide et al. (2021)	Volatility between oil shocks and metal markets	Causality test

Source: Compiled by the authors.

(Yadav et al., 2023)

ARCH model for volatility

- For stationary series, but with conditional variances that change over time.
- The ARCH (autoregressive conditional heteroskedastic) model captures conditional heteroskedasticity by modeling the variance of the error term as a function of past squared error terms.
- The error term is from a mean equation (e.g., ARMA, AR, or a simple constant model)

Covariance Stationary Series

- Mean is time-invariant
- Variance is constant
- All covariances are constant
 - all autocorrelations are constant

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Time-varying volatility: ARCH model

Suppose that instead of a conditional mean that changes over time we have a conditional variance that changes over time

- Consider a variant of the above model:

$$y_t = \phi + e_t$$

$$e_t | I_{t-1} \sim N(0, h_t)$$

“Shock”,
“Innovation”,
“News”

$$h_t = \alpha_0 + \alpha_1 e_{t-1}^2, \quad \alpha_0 > 0, \quad 0 \leq \alpha_1 < 1$$

Test ARCH effect

A Lagrange multiplier (LM) test is often used to test for the presence of ARCH effects

- To perform this test, first estimate the mean equation

$$y_t = \phi + e_t$$

$$\hat{e}_t^2 = \gamma_0 + \gamma_1 \hat{e}_{t-1}^2 + v_t$$

$$H_0 : \gamma_1 = 0 \quad H_1 : \gamma_1 \neq 0$$

The LM test statistic is $(T - q)R^2$

$$y_t = \phi + e_t$$

$$e_t | I_{t-1} \sim N(0, h_t)$$

$$\left(\frac{e_t}{\sqrt{h_t}} \middle| I_{t-1} \right) = z_t \sim N(0, 1)$$

$$h_t = \alpha_0 + \alpha_1 e_{t-1}^2, \quad \alpha_0 > 0, \quad 0 \leq \alpha_1 < 1$$

$$E(e_t) = E(z_t) E\left(\sqrt{\alpha_0 + \alpha_1 e_{t-1}^2}\right)$$

$$E(e_t^2) = E(z_t^2) E(\alpha_0 + \alpha_1 e_{t-1}^2) = \alpha_0 + \alpha_1 E(e_{t-1}^2)$$

GARCH model

The GARCH model extends ARCH by incorporating lagged conditional variances in addition to past squared errors.

An ARCH(q) model would be:

$$h_t = \alpha_0 + \alpha_1 e_{t-1}^2 + \alpha_2 e_{t-2}^2 \dots + \alpha_q e_{t-q}^2$$



GARCH(1,1)

$$h_t = \delta + \alpha_1 e_{t-1}^2 + \beta_1 h_{t-1}$$

$$\alpha_1 + \beta_1 < 1$$

1. Univariate GARCH Parameters (for each asset in the system)

- Omega (ω): The constant term in the variance equation.
- Alpha (α): Measures the impact of past shocks (ARCH effect).
- Beta (β): Measures the persistence of volatility (GARCH effect).

Optimal Parameters

	Estimate	Std. Error	t value	Pr(> t)
mu	0.034003	0.018302	1.8578	0.063192
omega	0.020800	0.004824	4.3113	0.000016
alpha1	0.081614	0.008655	9.4297	0.000000
beta1	0.910939	0.008937	101.9243	0.000000

$$\alpha + \beta \approx 1$$

- Highly persistent volatility
- Slow decay of shocks
- Long memory behavior

T-GARCH (Threshold GARCH)

T-GARCH is an extension of the GARCH model that accounts for asymmetric effects of shocks on volatility. It allows for the possibility that positive and negative shocks have different impacts on conditional variance.

$$h_t = \delta + \alpha_1 e_{t-1}^2 + \gamma d_{t-1} e_{t-1}^2 + \beta_1 h_{t-1}$$

$$d_t = \begin{cases} 1 & e_t < 0 \text{ (bad news)} \\ 0 & e_t \geq 0 \text{ (good news)} \end{cases}$$

Optimal Parameters

	Estimate	Std. Error	t value	Pr(> t)
mu	0.018964	0.018718	1.0131	0.310995
omega	0.018708	0.003998	4.6790	0.000003
alpha1	0.094526	0.009168	10.3106	0.000000
beta1	0.914457	0.008554	106.9060	0.000000
eta11	0.284468	0.052546	5.4137	0.000000

DCC-GARCH: testing time-varying conditional correlation

Dynamic Conditional Correlation (DCC-) GARCH

The idea of the models in this class is that the covariance matrix, H_t , can be decomposed into conditional standard deviations, D_t , and a correlation matrix, R_t . In the DCC-GARCH model both D_t and R_t are designed to be time-varying.

Suppose we have returns, a_t , from n assets with expected value 0 and covariance matrix H_t . Then the Dynamic Conditional Correlation (DCC-) GARCH model is defined as:

$$r_t = \mu_t + a_t$$

$$a_t = H_t^{1/2} z_t$$

$$H_t = D_t R_t D_t$$

DCC-GARCH

Mean equations for Y1 and Y2



1. Univariate GARCH(1,1) Process:

The conditional variances for ϵ_{1t} and ϵ_{2t} are:

$$\sigma_{1t}^2 = \omega_1 + \alpha_1 \epsilon_{1,t-1}^2 + \beta_1 \sigma_{1,t-1}^2$$

$$\sigma_{2t}^2 = \omega_2 + \alpha_2 \epsilon_{2,t-1}^2 + \beta_2 \sigma_{2,t-1}^2$$

Where:

- ω_1, ω_2 : Constant terms.
- α_1, α_2 : Impact of past squared residuals (ARCH effect).
- β_1, β_2 : Impact of past conditional variances (GARCH effect).



2. Dynamic Conditional Correlation (DCC):

The DCC model specifies the conditional correlation between ϵ_{1t} and ϵ_{2t} as time-varying:

$$\rho_{12,t} = \frac{q_{12,t}}{\sqrt{q_{11,t}q_{22,t}}}$$

Where:

- $\rho_{12,t}$: Conditional correlation between the two series at time t .
- $q_{11,t}, q_{22,t}$: Conditional variances of ϵ_{1t} and ϵ_{2t} , respectively.
- $q_{12,t}$: Conditional covariance between ϵ_{1t} and ϵ_{2t} .

[Joint]dcca1	0.019637	0.008256	2.3785	0.017382
[Joint]dccb1	0.939031	0.026749	35.1053	0.000000

3 Estimating Dynamic Conditional Correlations

Once the univariate GARCH models are fitted, the dynamic conditional correlations are modeled through a DCC process applied to the standardized residuals ($z_{i,t}$):

1. **Intermediate Correlation Matrix:** The model estimates an intermediate correlation matrix (Q_t) that evolves dynamically:

$$Q_t = (1 - a - b)\bar{Q} + a(z_{t-1}z_{t-1}^\top) + bQ_{t-1}$$

Here:

- Q_t : Time-varying covariance matrix,
- $\bar{Q}$: Unconditional correlation matrix of $z_{i,t}$,
- a : Coefficient capturing the short-term effect of past shocks on correlation,
- b : Coefficient capturing the persistence of past correlations,
- $z_t z_t^\top$: Outer product of standardized residuals at t .

Zhang (2015)

Example

Zhang, D. (2015). The trade effect of price risk: a system-wide approach. *Empirical Economics*, 48(3), 1149-1167.

In the GARCH method, the stochastic error is obtained from a moving-average AR(1) process or an autoregressive moving-average ARMA(1,1) process.¹³ For a typical AR(1)-GARCH(1,1) model, the conditional variance is obtained by jointly estimating the following equations:

$$\Delta \log p_t = b_0 + b_1 \Delta \log p_{t-1} + \mu_t \quad (14a)$$

$$v_t = a_0 + a_1 \mu_{t-1}^2 + a_2 v_{t-1}, \quad (14b)$$

Import demand under price and exchange-rate uncertainties: The case of US Atlantic salmon imports

D Zhang, Y Fang, Y Liu
Aquaculture 574, 739712

Profitability in Norwegian salmon farming: The impact of firm size and price variability

F Asche, M Sikveland, D Zhang
Aquaculture economics & management 22 (3), 306-317

Influence of price variability and financial ratios on business failure in the Atlantic Salmon industry

D Zhang, R Tveterås
Marine Resource Economics 37 (2), 183-200

Zheng et al. (2008)

Example

Zheng, Y., Kinnucan, H. W., & Thompson, H. (2008). News and volatility of food prices. *Applied Economics*, 40(13), 1629-1635.

Applied Economics, 2008, **40**, 1629–1635

News and volatility of food prices

Yuqing Zheng^a, Henry W. Kinnucan^b and Henry Thompson^{b,*}

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^b*Department of Agricultural Economics and Rural Sociology, Auburn University, Auburn, Alabama 36849, USA*

Financial markets exhibit an asymmetric news effect with unexpected low prices generating more price volatility than 'news' of high prices. The present study examines US food markets for such asymmetric news effects. Analysis of 25 years of monthly data for 45 retail food items shows that price news destabilizes about a third of the markets with unexpected price increases more destabilizing.

Zheng et al. (2008): EGARCH (Exponential GARCH)

the ARCH specification of Engle (1982) is

$$h_t = \omega + \sum_{i=1}^p \alpha_i \varepsilon_{t-i}^2 \quad (1)$$

where p is the number of lags and α_i and ω are parameters. An implicit assumption is that older news has less impact on current volatility than more recent news, $\alpha_i < \alpha_j$ for $i > j$. News that arrived more than p periods ago has no effect on current volatility.

The GARCH (p, q) model introduced by Bollerslev (1986) generalizes the ARCH (p) model by including the persistence term

$$h_t = \omega + \sum_{i=1}^p \alpha_i \varepsilon_{t-i}^2 + \sum_{i=1}^q \beta_i h_{t-i} \quad (2)$$

where q is the number of lags for the lagged variance term and the β_i are coefficients. This GARCH model is an infinite order ARCH model. Most empirical applications use the GARCH(1,1) implying the effect of the shock declines geometrically over time.

As noted by Engle and Ng (1993) a drawback of (1) and (2) is the implicit assumption that low-or high-price news has symmetric effects. The exponential GARCH or EGARCH model of Nelson (1991) relaxes this assumption,

$$\begin{aligned} \ln(h_t) = & \omega + \beta \ln(h_{t-1}) + \gamma \left(\frac{\varepsilon_{t-1}}{h_{t-1}^{1/2}} \right) \\ & + \alpha \left[\left(\frac{|\varepsilon_{t-1}|}{h_{t-1}^{1/2}} \right) - \left(\frac{2}{\pi} \right)^{1/2} \right] \end{aligned} \quad (3)$$

$$\ln(h_t) = \delta + \beta_1 \ln(h_{t-1}) + \alpha \left| \frac{e_{t-1}}{\sqrt{h_{t-1}}} \right| + \gamma \left(\frac{e_{t-1}}{\sqrt{h_{t-1}}} \right)$$

Zheng et al. (2008)

News and volatility of food prices

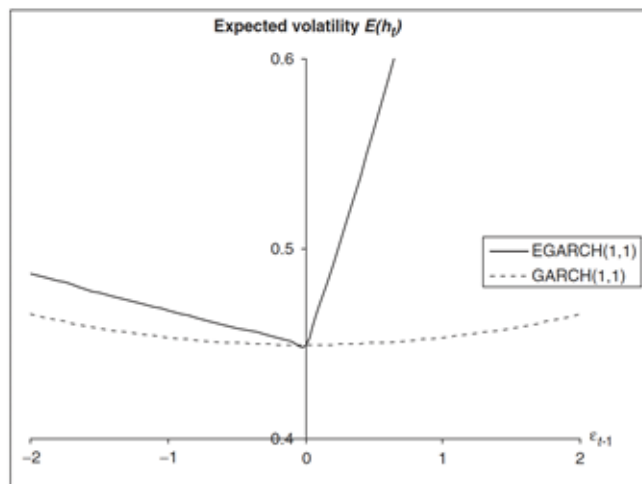


Fig. 1. The news impact curves of the GARCH(1, 1) model and EGARCH(1, 1) model

Notes: EGARCH(1,1) curve corresponds to equation:

$$\ln(h_t) = -0.8 + \ln(h_{t-1}) + 0.45(\varepsilon_{t-1}/h_{t-1}^{1/2}), h_{t-1} = 1, \quad \text{when } \varepsilon_{t-1} > 0$$

$$\ln(h_t) = -0.8 + \ln(h_{t-1}) + 0.04(\varepsilon_{t-1}/h_{t-1}^{1/2}), h_{t-1} = 1, \quad \text{when } \varepsilon_{t-1} < 0$$

GARCH(1,1) curve corresponds to equation:

$$h_t = -0.55 + h_{t-1} + 0.004(\varepsilon_{t-1}^2), h_{t-1} = 1$$

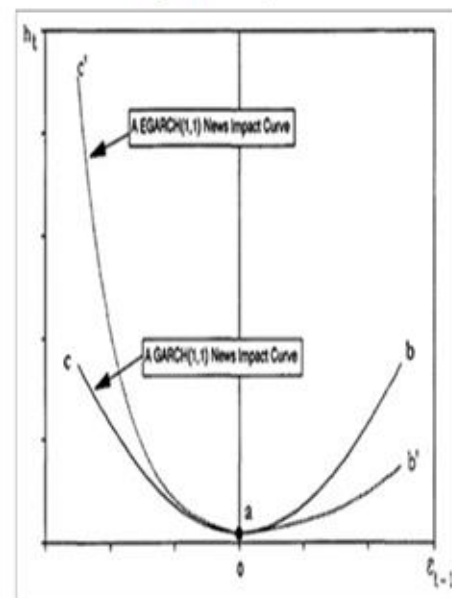
Ulusoy and Ördurak: The Impact of Oil Price Volatility to Oil and Gas Company Stock Returns and Emerging Economies

3. METHODOLOGY

exponential GARCH (EGARCH) instruments to volatility behavior of oil prices. Major advantage of at, instead of considering heteroskedasticity as a modeled. Usually financial data suggests that some : riskier than others; that is, the expected value of of error terms at sometimes is greater than at others. h models is to provide a volatility measure, like a ion, then can be used in financial decisions related is, portfolio selection and derivative pricing (Engle 01).

assumes that the variance of u_t in period t , σ_t^2 square of the error term in $t-1$ period, u_{t-1}

Figure 1: News impact curves



Zhang (2015)

Example

Zhang, D. (2015). Interdependence between Nordic stock markets and financial cooperation. *Review of accounting and Finance*, 14(2), 172-188.

Abstract

Purpose – This study aims to link the financial cooperation in the Nordic region and the interdependence between the stock markets in this area. The main emphasis is placed on the evolution of this interdependence as the financial integration was proceeding.

Design/methodology/approach – Johansen's cointegration technique and the exponential generalized autoregressive conditionally heteroskedastic model are applied to test the long-run and short-run interdependences, respectively, among Nordic stock markets. In particular, the recursive estimation approach is used to reveal the evolution of the interdependence between those markets.

Findings – The existence of two cointegrations over the sample period indicates that the markets depend on each other to some extent. The recursive estimation of Johansen's model further reveals that the interdependence had been greatly improving until late 2008. The interdependence between those markets is also confirmed convincingly by the short-term dynamics, noting that the spillover effects between most pairs of stock volatilities are witnessed in the empirical results.

Practical implications – The findings show the dynamics of the long-run correlations between the Nordic stock markets, which imply the intrinsic response to the process of financial market reforms, the 2008 global financial crisis and the period after the crisis. The evidenced information about determinants of the interdependence between Nordic stock markets is sending strong signals to investors to enhance their investment strategies.

Originality/value – Most of the existing studies have been restricted to the static long-run and/or short-run interdependence among those markets. However, this study contributes to the literature by investigating the dynamics of interdependence among the Nordic stock markets over time; moreover, the evolution of the market interdependence is sketched closely to the process of the regional financial market reforms.

Keywords Cointegration, Interdependence, EGARCH, Nordic region, Stock

1. Introduction

2. Literature review

3. Background and data

3.1 Background

3.1.1 Data

4. Methodological approach[1]

4.1 The cointegration method

4.2 Recursive estimation

4.3 EGARCH model

5. Empirical findings

5.1 The long-run structure

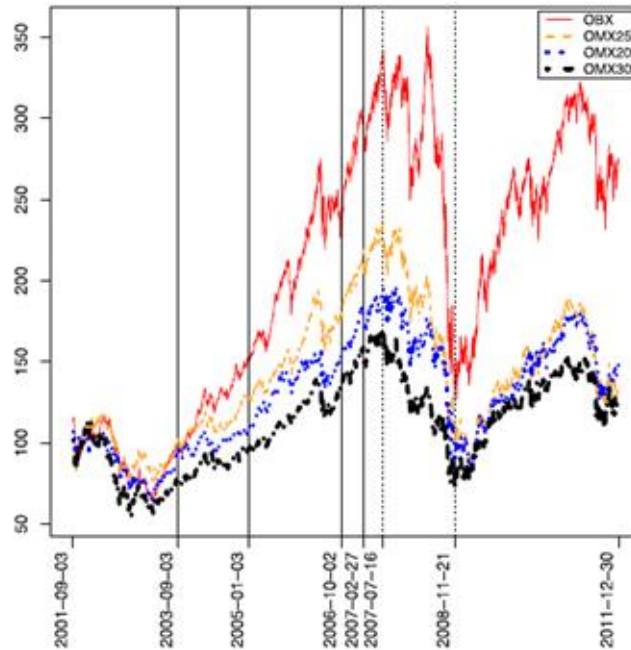
5.2 Recursive estimation of the VAR model

5.3 Exponential GARCH model

6. Concluding remarks

References

Zhang (2015)



Notes: The data are recalculated by taking Sep 2001 to Dec 200 as the base (100); the vertical full lines represent the important events of financial reforms in the Nordic market (Table I); the vertical broken lines range the period of the 2008 global finance crisis, which are approximately represented by the highest and lowest points of Sweden OMX25 during that period

Nordic stock
markets

177

Figure 1.
Nordic stock markets
indices (daily data
from Sep 2001 to Dec
2011)

Zhang (2015)

4. Methodological approach[1]

4.1 The cointegration method

In a multivariate time series context, the Johansen procedure (Johansen, 1991) has proven to be a useful tool to analyze the long-run relationship between closely related stock markets. The Johansen procedure is based on an unrestricted VAR represented by:

$$\Delta Y_t = \Pi Y_{t-1} + \Gamma_1 \Delta Y_{t-1} + \Gamma_2 \Delta Y_{t-2} + \dots + \Gamma_g \Delta Y_{t-g} + U_t \quad (1)$$

4.2 Recursive estimation

The aforementioned conventional cointegration approach does not consider the fact that the statistical relationship between markets may be a gradual and ongoing process (Serletis and King, 1997). One approach to reveal the dynamics of national markets is to estimate the time path of cointegrating space between index series. As the process of market interdependence deepens, the number of cointegrating vectors should increase to reflect the changes in the market (Rangvid, 2001). An alternative is to estimate the time-varying adjustment parameters [matrix α in equation (1)]. A larger value of the component in the matrix α indicates a faster speed of adjustment in response to the deviations from the long-run relationship ($\beta' \times Y_{t-1}$). The fast response to the disequilibrium should constitute evidence of increasing interdependence among the stock markets (Garcia Pascual, 2003).

The aforementioned approaches are both based on the recursive estimation methodology introduced by Hansen and Johansen (1999). For a sample with T observations, the baseline period is set to be 1: $(T_1 - 1)$, and the recursions are defined by adding more observations. Thus, the recursions are performed for the sub-samples 1: t , where $t = T_1, T_1 + 1 [\dots], T$. For each recursion, the statistic value of the trace test and the adjustment parameter are kept and then constitute two respective sets of time series, from which the degree of capital market interdependence can be assessed. Because the trace test is based on an eigenvalue, which is a quadratic function of α and β (Juselius, 2006), consistency between the two approaches is theoretically expected.

Zhang (2015)

assumption of the exogeneity of some of the indices is not correct. Therefore, we apply a four-dimensional VAR-EGARCH model confined to Denmark, Norway, Sweden and Finland, which is represented by:

$$R_{it} = \beta_{i0} + \sum_{j=1}^4 \beta_{ij} R_{jt-1} + \varepsilon_{it} \quad \text{for } i = 1, 2, 3, 4; \quad (2)$$

$$\sigma_{it}^2 = \exp \left\{ \alpha_{i0} + \sum_{j=1}^4 \alpha_{ij} f_j(z_{jt-1}) + \gamma_i \ln(\sigma_{it-1}^2) \right\} \quad \text{for } i = 1, 2, 3, 4; \quad (3)$$

$$f_j(z_{jt-1}) = (|z_{jt-1}| - E|z_{jt-1}|) + \delta_j z_{jt-1} \quad \text{for } j = 1, 2, 3, 4; \quad (4)$$

	Denmark	Norway	Sweden	Finland
<i>Price spillover parameters</i>				
β_{i0}	0.0521** (0.0203)	0.0782** (0.0258)	0.0297 (0.0229)	0.0416** (0.0211)
β_{i1}	-0.0496* (0.0285)	-0.0614* (0.0317)	-0.0169 (0.0315)	-0.0399 (0.0291)
β_{i2}	-0.010 (0.0201)	-0.0331 (0.0233)	-0.0292 (0.0195)	0.0043 (0.0198)
β_{i3}	0.0724** (0.0301)	0.1266** (0.0294)	0.0070 (0.0319)	0.0841** (0.0302)
β_{i4}	0.0227 (0.0332)	-0.0127 (0.028)	0.0025 (0.0326)	-0.0129 (0.0324)
<i>Volatility spillover parameters</i>				
α_{i0}	0.0110** (0.003)	0.0200** (0.0045)	0.0111** (0.0024)	0.0106** (0.0019)
α_{i1}	0.0891** (0.0154)	0.0456** (0.0176)	0.0524** (0.0118)	0.0596** (0.0101)
α_{i2}	0.0318** (0.0154)	0.1062** (0.0202)	-0.0171 (0.0113)	0.0159* (0.0089)
α_{i3}	0.0514** (0.0137)	0.0256** (0.0123)	0.0772** (0.0149)	0.0399** (0.0101)
α_{i4}	-0.0640** (0.0157)	-0.0796** (0.0153)	-0.0502** (0.0134)	-0.0255** (0.0103)
<i>Other parameters</i>				
γ_i	0.9680** (0.005)	0.9725** (0.0052)	0.9815** (0.0026)	0.9809** (0.0025)
δ_i	0.0166 (0.1401)	-0.0987 (0.1149)	-0.6366** (0.2129)	0.5266** (0.1773)
<i>Correlation matrix</i>				
Denmark	1	0.6284** (0.0103)	0.6726** (0.0099)	0.6935** (0.0093)
Norway		1	0.6702** (0.0104)	0.7238** (0.0087)
Sweden			1	0.8075**

Why are asymmetric ARCH models important?

price news has symmetric effects. The exponential GARCH or EGARCH model of Nelson (1991) relaxes this assumption,

$$\ln(h_t) = \omega + \beta \ln(h_{t-1}) + \gamma \left(\frac{\varepsilon_{t-1}}{h_{t-1}^{1/2}} \right) + \alpha \left[\left(\frac{|\varepsilon_{t-1}|}{h_{t-1}^{1/2}} \right) - \left(\frac{2}{\pi} \right)^{1/2} \right] \quad (3)$$

$$h_t = \delta + \alpha_1 e_{t-1}^2 + \gamma d_{t-1} e_{t-1}^2 + \beta_1 h_{t-1}$$

$$d_t = \begin{cases} 1 & e_t < 0 \text{ (bad news)} \\ 0 & e_t \geq 0 \text{ (good news)} \end{cases}$$

Oil Prices and Stock Markets: A Review of the Theory and Empirical Evidence

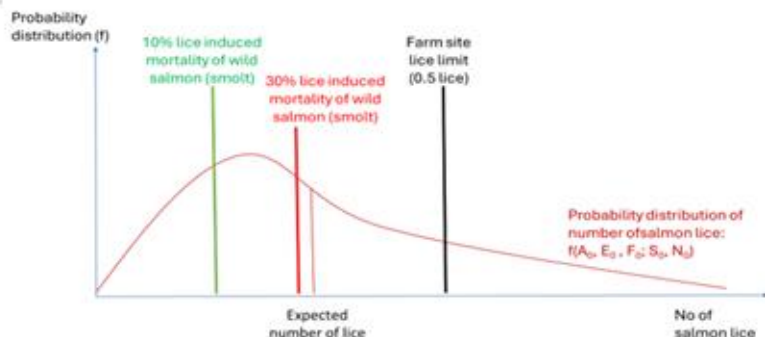
Stavros Degiannakis, George Filis,** and Vipin Arora****

ABSTRACT

Do oil prices and stock markets move in tandem or in opposite directions? The complex and time varying relationship between oil prices and stock markets has caught the attention of the financial press, investors, policymakers, researchers, and the general public in recent years. In light of such attention, this paper reviews research on the oil price and stock market relationship. The majority of papers we survey study the impacts of oil markets on stock markets, whereas, little research in the reverse direction exists. Our review finds that the causal effects between oil and stock markets depend heavily on whether research is performed using aggregate stock market indices, sectorial indices, or firm-level data and whether stock markets operate in net oil-importing or net oil-exporting countries. Additionally, conclusions vary depending on whether studies use symmetric or asymmetric changes in the price of oil, or whether they focus on unexpected changes in oil

“Myopic loss aversion (MLA) is a behavioral economics concept where investors, driven by the tendency to fear losses more than enjoy equivalent gains, focus too frequently on short-term portfolio volatility.”

Implikasjoner av reguleringsmodeller for norsk havbruksnæring og samfunnet (HAVREG)



1. Mean Function (Expected Lice Count) without production area, farm or firm effect:

$$E(L_i) = f(N_i, T_i, E_i, F_i, S_i; \beta)$$

|

2. Variance Function (Heteroskedastic Risk)

$$\text{Var}(L_i) = \sigma^2(N_i, T_i, E_i, F_i, S_i; \gamma)$$

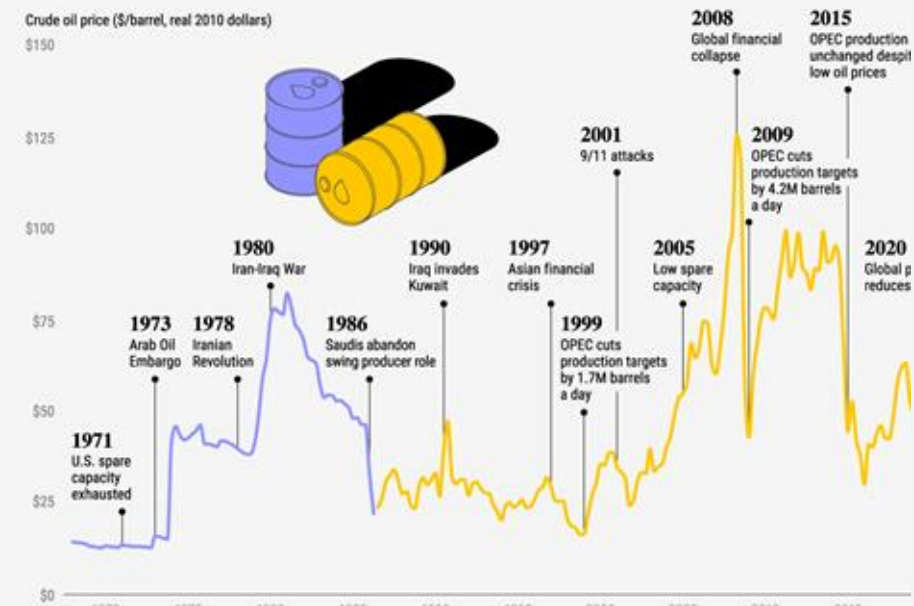
Part 2: Research topics and theories

Research topics

For the returns of the oil price (**Oil-Price**) and stock prices of a traditional oil company (**Oil-Company**) and a renewable energy company (**Renewable-Energy**):

- For Oil-Company and Renewable-Energy
 - ARCH
 - GARCH
 - T-GARCH
- Oil-Company vs. Oil-Price
 - DCC-GARCH
- Renewable-Energy vs. Oil-Price
 - DCC-GARCH

$$\ln\left(\frac{P_t}{P_{t-1}}\right) \times 100$$



Source: U.S. Energy Information Administration, Refinitiv (Mar 2022)

C. Oil price volatility and key events



Brent Spot



United States Oil Fund, LP (USO)

119.89 +1.50 (+1.27%)

118.25 -1.64 (-1.37%)

☆ Follow

NYSEArca - Nasdaq Real Time Price - USD

At close: March 13 at 4:00:00 PM EDT

After hours: March 13 at 7:59:59 PM EDT

Mount

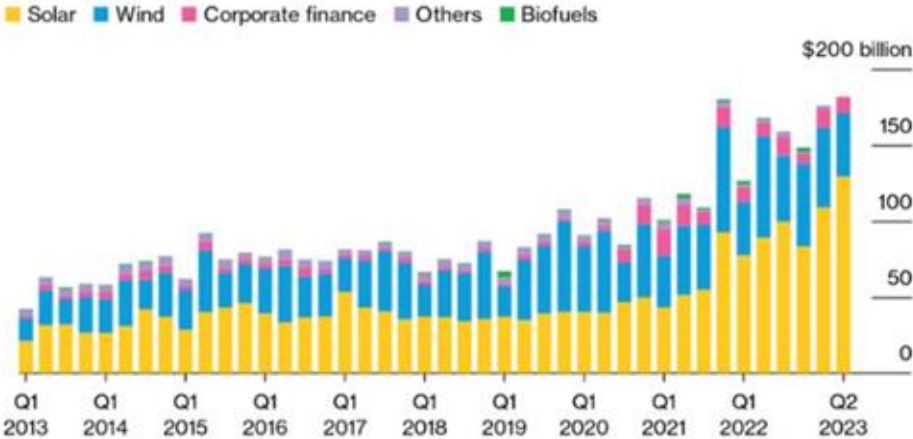
O:71.59 H:76.97 L:71.45 C:74.98 V:19.2m

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Renewable Energy Investment Hit Record \$358 Billion in 1H 2023

Global quarterly investment in renewable energy

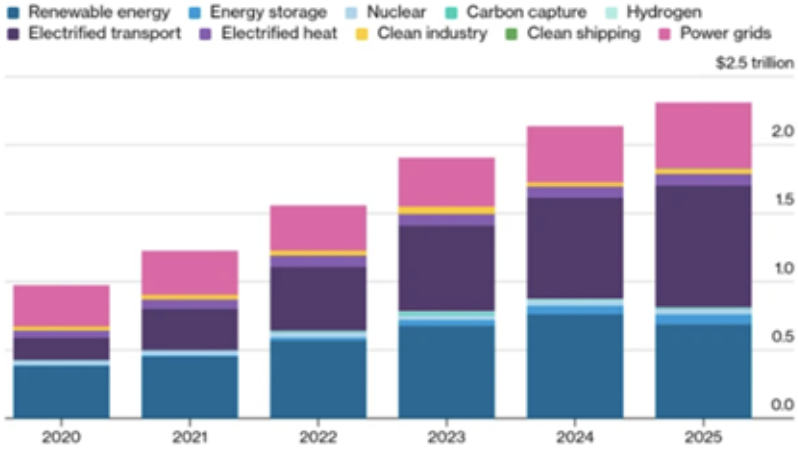


Source: BloombergNEF

BloombergNEF

Energy Transition Investment Reaches \$2.3 Trillion in 2025

Global energy transition investment by sector, 2020-25



Source: BloombergNEF

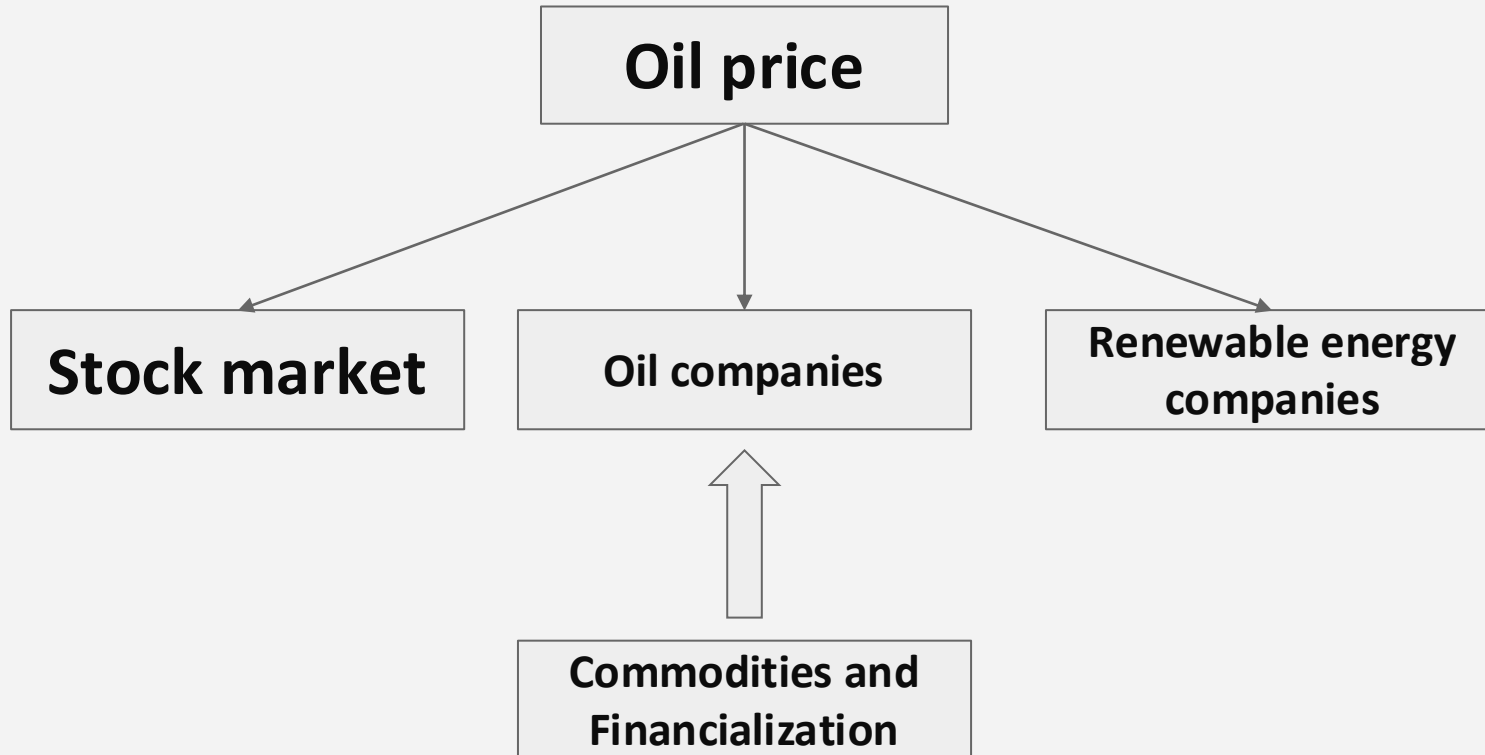
BloombergNEF

Volatility in stock markets in general

- Le, T. P. T. D., & Tran, H. L. M. (2021). The contagion effect from US stock market to the Vietnamese and the Philippine stock markets: The evidence of DCC-GARCH model. *The Journal of Asian Finance, Economics and Business*, 8(2), 759-770.
- Lim, D., Goh, K. W., & Sim, Y. W. (2023). Estimation of stock market index volatility using the GARCH model: Causality between stock indices. *Asian Economic and Financial Review*, 13(3), 162-179.
- Zhang, D. (2015). Interdependence between Nordic stock markets and financial cooperation. *Review of accounting and Finance*, 14(2), 172-188.

- ❖ Financial volatilities can arise due to increasing in inflationary expectations, restricting monetary policy, interest rate change, and economic crises (Le & Tran, 2021).
- ❖ Contagion happens when the cross-market correlation goes up significantly in the period of crisis.
- ❖ Volatility spillover vs. price spillover
 - No feedback between price spillovers; the volatility spillover creates a feedback effect between Sweden and Finland (Booth et al. 1997)
- ❖ Investors can forecast stock market index volatility to set strategies for risk management and portfolio optimization (Lim et al., 2015)

Theoretical framework



Oil price and stock markets, search:

<https://scholar.google.com/schhp?hl=en>

oil price and stock market

About 2 290 000 results (0,15 sec)

10 results loaded 1 of which are from potentially predatory journals

The impact of oil price shocks on the US stock market

Q1 4 L Kilian, C Park - International economic review, 2009 - Wiley Online Library

... of the link between **oil prices** and **stock** returns. Second, our ... in **oil prices** in recent years has not caused a **stock market** ... US **stock** returns to unanticipated changes in the **price of oil** are ...

☆ Save Cite Cited by 2793 Related articles All 11 versions

Oil price shocks, stock market, economic activity and employment in Greece

Q1 3 E Panapetrou - Energy economics, 2001 - Elsevier

... dynamic relationship among **oil prices**, real **stock prices**, interest rates, ... **oil price** changes affect real economic activity and employment. **Oil prices** are important in explaining **stock price** ...

☆ Save Cite Cited by 1263 Related articles All 11 versions

Relationships between oil price shocks and stock market: An empirical analysis from China

Q1 2 RG Cong, YM Wei, JL Jiao, Y Fan - Energy Policy, 2008 - Elsevier

... **oil price** shocks and Chinese **stock market** using multivariate vector auto-regression. **Oil price** ... significant impact on the real **stock** returns of most Chinese **stock market** indices, except for ...

☆ Save Cite Cited by 993 Related articles All 16 versions

Oil price shocks and stock market activities: Evidence from oil-importing and oil-exporting countries

Q1 3 Y Wang, C Wu, L Yang - Journal of Comparative economics, 2013 - Elsevier

... response by **stock market** in a country to **oil price** shocks highly ... in the world **oil market**, and whether changes in **oil price** are ... of **oil price** shocks depends on the level of importance of **oil** ...

☆ Save Cite Cited by 788 Related articles All 8 versions

Degiannakis, S., Filis, G., & Arora, V. (2018). Oil prices and stock markets: A review of the theory and empirical evidence. *The Energy Journal*, 39(5), 85-130.

Oil price vs stock indices or public firms

“Our review finds that the causal effects between oil and stock markets depend heavily on whether research is performed using:

- aggregate stock market indices,
- sectoral indices, or
- firm-level data, and
- whether stock markets operate in net oil-importing or net oil-exporting countries.”

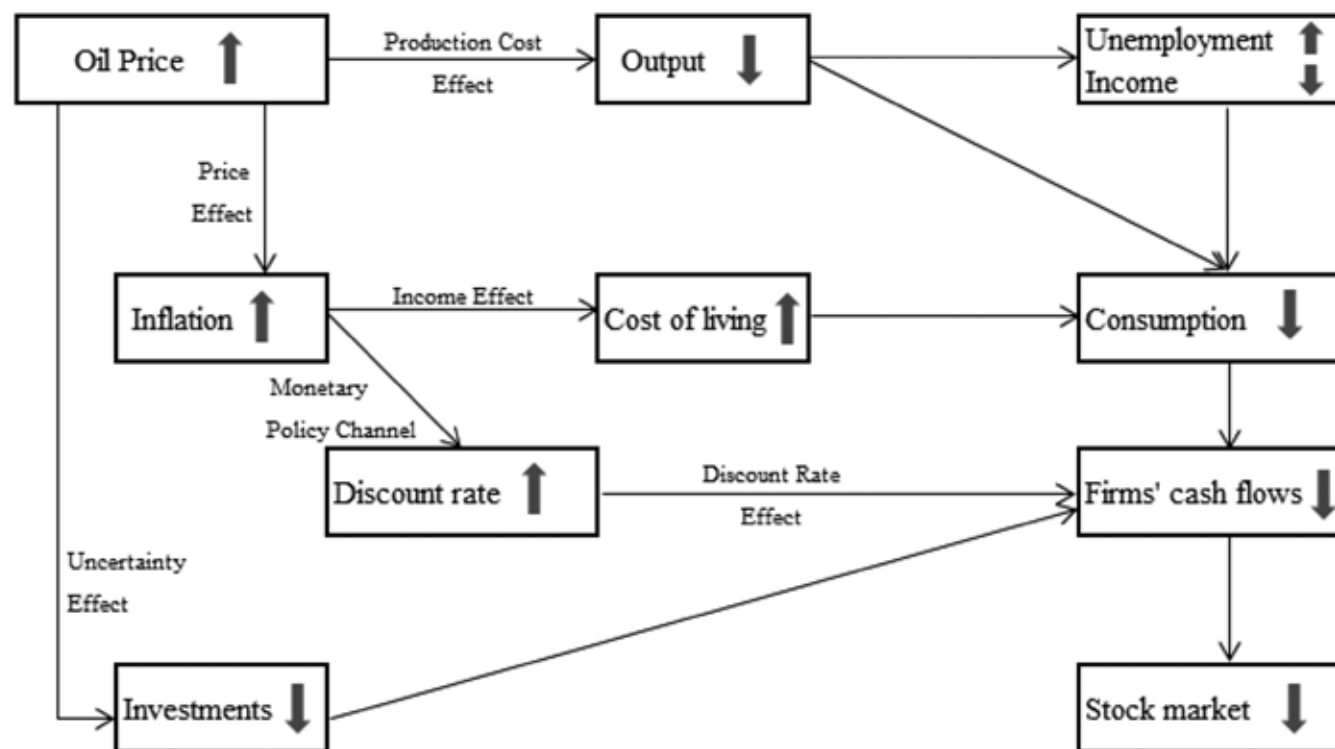
(Degiannakis et al., 2018)

Oil price and stock markets: the theory

- Stock valuation channel
- Monetary channel
- Output channel
- Fiscal channel
- Uncertainty channel

(Degiannakis et al. 2018)

Figure 7: Transmission channel of positive oil price changes.



Adapted from Tang et al. (2010).

Oil price and oil companies

- Antonakakis, N., Cunado, J., Filis, G., Gabauer, D., & De Gracia, F. P. (2018). Oil volatility, oil and gas firms and portfolio diversification. *Energy Economics*, 70, 499-515.
- Sedighi, M., Mohammadi, M., Farahani Fard, S., & Sedighi, M. (2019). The nexus between stock returns of oil companies and oil price fluctuations after heavy oil upgrading: Toward theoretical Progress. *Economies*, 7(3), 71.
- Ulusoy, V., & Özdurak, C. (2018). The impact of oil price volatility to oil and gas company stock returns and emerging economies. *International Journal of Energy Economics and Policy*, 8(1), 144-158.

- ❖ Crude oil price affects financial markets and economic systems.
- ❖ Crude oil prices are volatile due to **supply, demand**, derivatives, political events, and crises (Sedighi et al., 2019).
- ❖ The fluctuations in oil prices are reflected in financial markets, particularly in the stock exchange.
- ❖ Aggregate and sectoral indices may mask the heterogeneity at the firm level.
- ❖ “The spillover is usually **unidirectional** from oil and gas companies’ stock volatility to oil price volatility” (Antonakakis et al., 2018, p. 499).

Oil price and renewable energy companies

- Henriques, I., & Sadorsky, P. (2008). Oil prices and the stock prices of alternative energy companies. *Energy Economics*, 30(3), 998-1010.
- Sadorsky, P. (2012). Correlations and volatility spillovers between oil prices and the stock prices of clean energy and technology companies. *Energy Economics*, 34(1), 248-255.

❖ Clean energy stock prices are influenced by technology stocks, oil prices, and probably carbon prices.

“In the long run, households and businesses respond to higher fuel prices by cutting consumption, purchasing products that are more efficient, and **switching to alternative energy sources**. Higher energy prices also encourage entrepreneurs to invest in the research and development of new energy-conserving technologies and alternative fuels, further expanding the opportunities available to households and businesses to reduce energy use and switch to low-cost sources”. Economic Report of the President (2006, page 243).

Commodity price and stock markets

As summarized in Zhang, Myrland & Xie (2016):

- Recent research typically focuses on the impact of commodity financialization on commodity prices and the impact of commodity prices on the stock market (Falkowski, 2011).
- The large increase in commodity price volatilities can attribute partially to commodities' financialization process (Tang & Xiong, 2012).
- Over a long horizon, both stock and commodity prices are expected to move in a similar direction as both respond to future economic performance (Black et al., 2014).

Do you think the ARCH models are useful for investment?

Read: Sadorsky, P. (2012). Correlations and volatility spillovers between oil prices and the stock prices of clean energy and technology companies. Energy economics, 34(1), 248-255.

Practical implications for investors

6. Portfolio weights

The conditional volatilities from MGARCH models can be used to construct optimal portfolio weights (Kroner and Ng, 1998).

$$w_{ij,t} = \frac{h_{jj,t} - h_{ij,t}}{h_{ii,t} - 2h_{ij,t} + h_{jj,t}}$$

$$w_{ij,t} = \begin{cases} 0, & \text{if } w_{ij,t} < 0 \\ w_{ij,t}, & \text{if } 0 \leq w_{ij,t} \leq 1 \\ 1, & \text{if } w_{ij,t} > 1 \end{cases}$$

In constructing portfolio weights between two assets, $w_{ij,t}$ is the weight of the first asset in a one dollar portfolio of two assets (asset i , asset j) at time t , $h_{ij,t}$ is the conditional covariance between assets i and j and $h_{jj,t}$ is the conditional variance of asset j . The weight of the second asset is $1 - w_{ij,t}$. Summary statistics for portfolio weights computed from the DCC model are reported in Table 7. The average weight for

5. Hedging

The conditional volatility estimates can be used to construct hedge ratios (Kroner and Sultan, 1993). A long position in one asset (say asset i) can be hedged with a short position in a second asset (say asset j). The hedge ratio between asset i and asset j is

$$\beta_{ijt} = h_{ijt} / h_{jtt}$$

For most of the hedge ratios, computed from the DCC model, the graphs show considerable variability after August 2008 (Fig. 4). For

Part 3: Methods and Previous studies

List of references

- References in Part 2 (about energy companies)
 - Henriques & Sadorsky (2008)
 - Sadorsky (2012)
 - Antonakakis et al. (2018)
 - Ulusoy & Özdurak (2018).
- Methods references (not about energy companies):
 - ARCH
 - GARCH: Lim et al. (2023)
 - T-GARCH
 - DCC-GARCH: Yadav et al. (2023); Le and Tran (2021).
- Le, T. P. T. D., & Tran, H. L. M. (2021). The contagion effect from US stock market to the Vietnamese and the Philippine stock markets: The evidence of DCC-GARCH model. *The Journal of Asian Finance, Economics and Business*, 8(2), 759-770.
- Lim, D., Goh, K. W., & Sim, Y. W. (2023). Estimation of stock market index volatility using the GARCH model: Causality between stock indices. *Asian Economic and Financial Review*, 13(3), 162-179.
- Yadav, N., Singh, A. B., & Tandon, P. (2023). Volatility spillover effects between Indian stock market and global stock markets: A DCC-GARCH model. *FIIIB Business Review*, 23197145221141186.

References in Part 2

Henriques & Sadorsky (2008)

The VAR model consists of four variables, the WilderHill Clean Energy Index (ECO), the Arca Technology Index (PSE), U.S. West Texas Intermediate crude oil futures prices (OIL), and the interest rate (RATE). To reduce unwanted variability (heteroskedasticity) in the data, natural logarithms are taken of each data series (LECO, LPSE, LOIL, LRATE).

A vector autoregression can be written in the following way.

$$y_t = \sum_{j=1}^p B_j y_{t-j} + u_t,$$

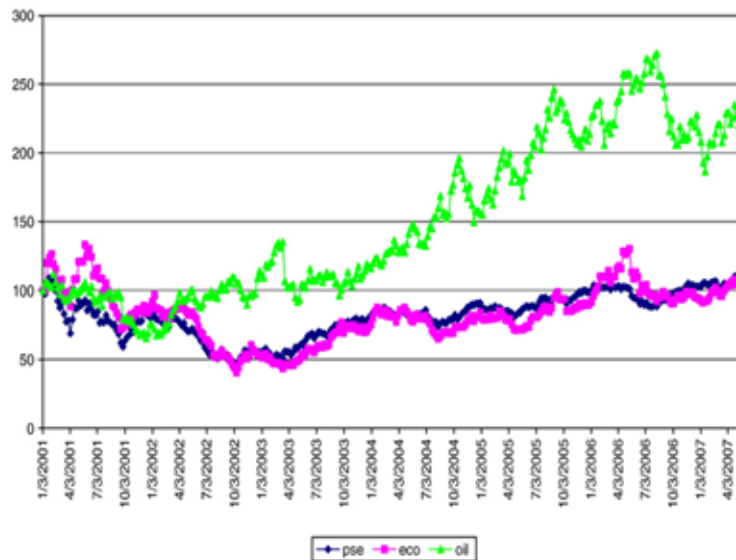


Fig. 1. ECO, PSE, and oil prices.

References in Part 2

Henriques & Sadorsky (2008)

Table 6

Toda and Yamamoto (1995) LA-VAR Wald tests

Dependent variable

	LECO	LPSE	LOIL	LRATE
LECO	–	7.833	3.863	7.283
LPSE	26.140***	–	8.559	8.821
LOIL	18.457***	16.068**	–	11.310
LRATE	21.004***	30.133***	17.280**	–

***, **, *, denote a test statistic is statistically significant at the 1%, 5%, or 10% level of significance.

References in Part 2

Sadorsky (2012)

In Eq. (1) r_{it} is the return for series i and ε_{it} is random error term with conditional variance h_{it} . The market information available at time $t - 1$ is denoted as I_{it-1} . Eq. (2) specifies the relation between the error term ε_{it} and the conditional variance h_{it} . Eq. (3) specifies a GARCH(1,1) process with VARMA terms (Ling and McAleer, 2003). The Ling and McAleer (2003) approach to modeling the conditional variances allows large shocks to one variable to affect the variances of the other variables. This is a convenient specification which allows for volatility spillovers.

The Engle (2002) dynamic conditional correlation (DCC) model is estimated in two steps. In the first step, the GARCH parameters are estimated. In the second step, the correlations are estimated.

$$H_t = D_t R_t D_t \quad (4)$$

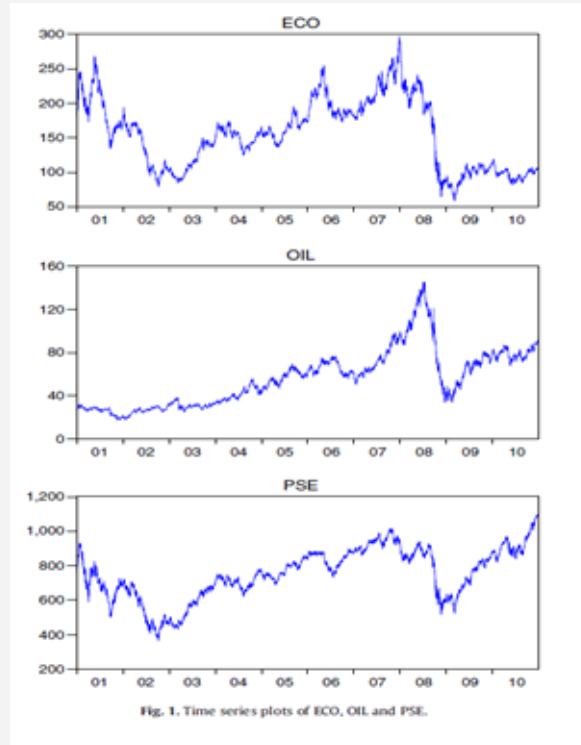
In Eq. (4), H_t is the 3×3 conditional covariance matrix, R_t is the conditional correlation matrix, and D_t is a diagonal matrix with time-varying standard deviations on the diagonal.

$$D_t = \text{diag}(h_{11t}^{1/2}, \dots, h_{33t}^{1/2}) \quad (5)$$

$$R_t = \text{diag}(q_{11t}^{-1/2}, \dots, q_{33t}^{-1/2}) Q_t \text{diag}(q_{11t}^{-1/2}, \dots, q_{33t}^{-1/2}) \quad (6)$$

Q_t is a symmetric positive definite matrix.

$$Q_t = (1 - \theta_1 - \theta_2) \bar{Q} + \theta_1 \xi_{t-1} \xi'_{t-1} + \theta_2 Q_{t-1} \quad (7)$$



References in Part 2

Sadorsky (2012)

Table 4
MGARCH parameter estimates

	Mean		DCC		
			Coeff	T stat	Signif
	m_{10}		0.040	1.085	0.278
	m_{11}		0.016	0.464	0.643
	m_{12}		0.025	1.333	0.183
	m_{13}		0.069	1.668	0.095
	m_{20}		0.098	2.129	0.033
	m_{21}		-0.014	-0.456	0.648
	m_{22}		-0.019	-0.838	0.402
	m_{23}		0.053	1.310	0.190
	m_{30}		0.061	2.530	0.011
	m_{31}		-0.024	-1.052	0.293
	m_{32}		0.007	0.512	0.609
	m_{33}		-0.018	-0.552	0.581
Variance	c_{11}		0.084	2.920	0.004
	c_{21}		0.102	2.197	0.028
	c_{22}				
	c_{31}				
	c_{32}				
	c_{33}		0.015	2.998	0.003
	α_{11}		0.080	2.587	0.010
	α_{12}		0.020	1.805	0.071
	α_{13}		-0.011	-0.256	0.798
	α_{21}		0.048	1.602	0.109
	α_{22}		0.057	3.566	0.000
	α_{23}		-0.010	-0.186	0.853
	α_{31}		0.021	1.925	0.054
	α_{32}		0.005	0.934	0.350
	α_{33}		0.026	2.423	0.015
	β_{11}		0.864	15.884	0.000
	β_{12}		-0.006	-0.247	0.805
	β_{13}		0.065	1.003	0.316
	β_{21}		-0.071	-1.759	0.079
	β_{22}		0.930	45.195	0.000
	β_{23}		0.009	0.137	0.891
	β_{31}		-0.034	-2.295	0.022
	β_{32}		-0.006	-0.546	0.585
	β_{33}		0.980	72.821	0.000
	ρ_{21}				
	ρ_{31}				
	ρ_{32}				
	θ_1		0.024	4.341	0.000
	θ_2		0.972	129.913	0.000
Log L			-142.364		
AIC			10.944		
SIC			11.023		

For the DCC model, the estimated coefficients on θ_1 and θ_2 are each positive and statistically significant at the 1% level. These estimated coefficients sum to a value which is less than one, meaning that the dynamic conditional correlations are mean reverting.

References in Part 2

Antonakakis et al. (2018)

Table 1
Descriptive statistics of oil and stock price volatility (absolute returns).

	Mean	Median	Max	Min	Std. Dev.	Skewness	Kur
WTI	1.683	1.232	17.092	0.000	1.717	2.510	13.4
BP	1.186	0.813	17.192	0.000	1.338	3.644	27.4
CHEVRON	1.109	0.818	18.941	0.000	1.177	3.829	34.2
CNPC	1.700	1.157	18.698	0.000	1.874	2.789	15.2
ENI	1.289	0.920	13.913	0.000	1.353	2.676	14.2
EXXON	1.038	0.751	15.863	0.000	1.130	3.791	31.2
LUKOIL	1.791	1.221	39.502	0.000	2.099	4.416	46.2
PETROBRAS	2.169	1.591	23.679	0.000	2.227	2.656	15.2
SHELL	1.150	0.829	15.721	0.000	1.257	3.296	22.2
SINOPEC	2.170	1.512	19.190	0.000	2.271	2.276	10.2
STATOIL	1.560	1.176	16.128	0.000	1.574	2.558	14.0
TOTAL	1.236	0.874	13.641	0.000	1.278	2.728	16.2
VALERO	1.807	1.350	22.314	0.000	1.811	2.845	18.0

Notes: * denotes significance at the 1% level.

3.2.1. Dynamic conditional correlation

To examine the time-varying correlations in the volatilities of each of the major oil and gas companies' stock prices and of oil prices we employ the DCC model proposed by Engle (2002).

The DCC model of Engle (2002) is defined as:

$$r_t = \mu_t(\theta) + \epsilon_t, \quad \text{where } \epsilon_t | \Omega_{t-1} \sim N(0, H_t) \quad (1)$$

$$\epsilon_t = H_t^{1/2} u_t, \quad \text{where } u_t \sim N(0, I) \quad (2)$$

$$H_t = D_t R_t D_t \quad (3)$$

where $r_t = (r_{1t}, \dots, r_{Nt})'$ is a $N \times 1$ vector of volatilities (specifically, the WTI, BP, CHEVRON, CNPC, ENI, EXXON, LUKOIL, PETROBRAS, SHELL, SINOPEC, STATOIL, TOTAL and the VALERO volatilities, thus $N = 13$); $\mu_t(\theta) = (\mu_{1t}, \dots, \mu_{Nt})'$ is the conditional 13×1 mean vector of r_t , H_t is the conditional covariance matrix, $D_t = \text{diag}(h_{1t}^{1/2}, \dots, h_{Nt}^{1/2})'$ is a diagonal matrix of square root conditional variances, where h_{it} can be defined as any univariate GARCH-type model, and R_t is the $t \times \left(\frac{N(N-1)}{2}\right)$ matrix containing the time-varying conditional correlations defined as:

$$R_t = \text{diag}(q_{11,t}^{-1/2}, \dots, q_{NN,t}^{-1/2}) Q_t \text{diag}(q_{11,t}^{-1/2}, \dots, q_{NN,t}^{-1/2}) \quad \text{or} \quad (4)$$

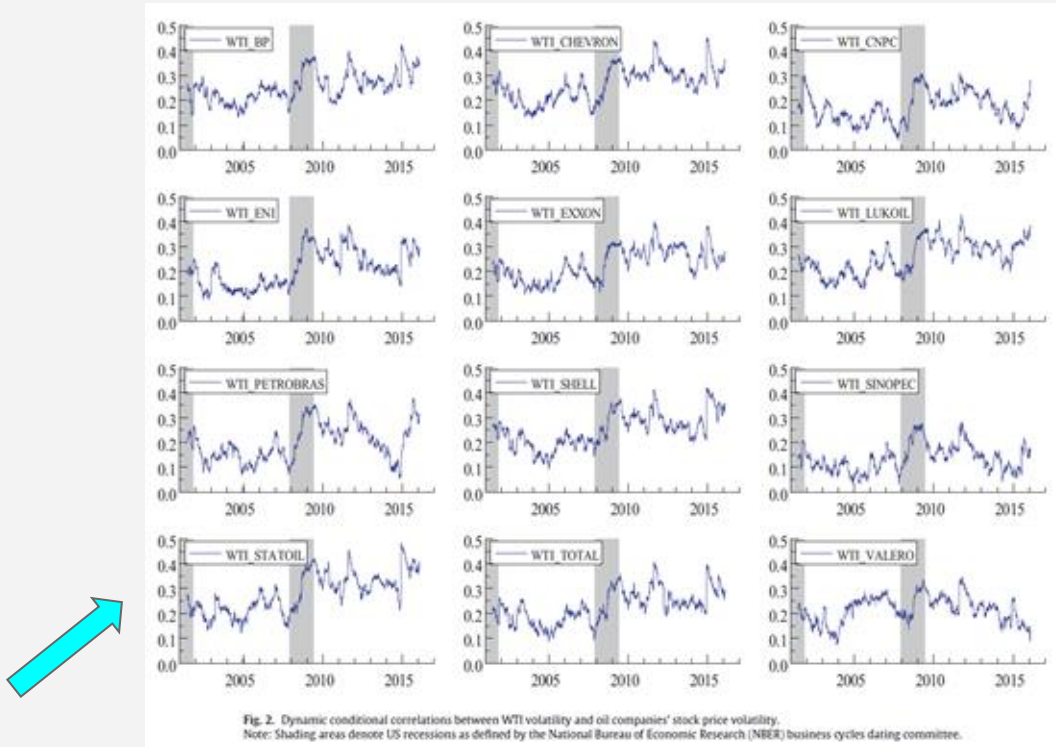
$$\rho_{ij,t} = \rho_{ji,t} = \frac{q_{ij,t}}{\sqrt{q_{ii,t}q_{jj,t}}}$$

where $Q_t = (q_{ij,t})$ is a $N \times N$ symmetric positive definite matrix given by:

$$Q_t = (1 - \alpha - \beta)\bar{Q} + \alpha u_{t-1}u_{t-1}' + \beta Q_{t-1} \quad (5)$$

References in Part 2

Antonakakis et al. (2018)



References in Part 2

Ulusoy, V., & Özdurak, C. (2018). The impact of oil price volatility to oil and gas company stock returns and emerging economies. *International Journal of Energy Economics and Policy*, 8(1), 144-158.

ARCH model assumes that the variance of u_t in period t , σ_t^2 depends on the square of the error term in $t-1$ period, u_{t-1} .

In this context, ARCH(q) and GARCH(q) models are as follows;

$$\alpha_0 > 0, \alpha_i > 0$$

$$h_t = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \alpha_2 \varepsilon_{t-2}^2 + \dots + \alpha_q \varepsilon_{t-q}^2 + v_t \quad (1)$$

GARCH models which express the generalized form of ARCH models were developed by Engle (1982) and Bollerslev (1986) to provide reliable estimations and predictions. GARCH models consist of conditional variance, in equation (2) in addition to conditional mean in equation (1).

$$h_t = \alpha_0 + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^p \beta_j h_{t-j} \quad (2)$$

In this context, restrictions of variance model are as follows:

$$\text{for } \alpha_i \geq 0 \text{ and } \beta_i \geq 0, \alpha_i + \beta_i < 1$$

Table 2: RWTI GARCH model for overall period

Distribution	Normal distribution				Student t-distribution			
	Mean equation		Variance equation		Mean equation		Variance equation	
	Coefficient	z-Stats	Coefficient	z-Stats	Coefficient	z-Stats	Coefficient	z-Stats
RWTI(-1)	-0.04	-2.69			-0.04	-3.17		
RCAD	-0.88	-14.29			-0.89	-14.67		
REUR	-0.52	-5.22			-0.40	-3.44		
RCHF	0.00	0.00			0.08	1.27		
RGBP	0.04	0.65			0.08	1.20		
RJPY	0.29	6.66			0.25	5.32		
RDX	-0.87	-6.95			-0.74	-4.84		
RMXN	-0.09	-2.02			-0.12	-2.73		
RRUB	-0.66	-21.75			-0.66	-18.28		
α_0			0.00	4.61				
α_1			0.07	14.00			0.06	8.31
β_1			0.92	155.23			0.94	132.1
Observations				4266				4266
R ²				0.177				0.178
DW				2.029				2.087

Methods – EGARCH: Lim et al. (2023)

Estimation of stock market index volatility using the GARCH model: Causality between stock indices



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(+ Corresponding author)

ABSTRACT

This paper aims to model the volatility of returns for selected stock indices and examine the causal relationships between the markets using the historical daily prices of the Financial Times Stock Exchange (FTSE), Bursa Malaysia Kuala Lumpur Composite Index (KLCI), the Indonesia Stock Exchange Index (LQ45), and the Stock Exchange of Thailand (SET) from January 2008 to November 2019. The study employs univariate GARCH models that are prominent in capturing the volatility clustering of financial instruments in association with the Box–Jenkins methodology for better estimation. Generally, the ARMA–GARCH model is used to capture the volatility series, while the Granger causality test examines the causal directions between the markets. The findings revealed leverage effects on the markets, with the outperformance of the EGARCH in analyzing the empirical properties of stock returns. An initial test that yielded positive correlations suggests the existence of co-movement between the derived volatility series. The study concluded bidirectional causal relationships between the selected markets, and based on the resulting relationships, it is proposed that supervision of markets among the ASEAN members could be advantageous in predicting the corresponding market performance.

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Keywords

ASEAN
Causality
Emerging market
Stock market
Univariate GARCH
Volatility estimation.

JEL Classification:

O1A, O17.

Methods – EGARCH: Lim et al. (2023)

The simplest GARCH used is known as GARCH(1,1), while higher order GARCH models are denoted as GARCH(p,q), where p indicates the order of the GARCH terms σ^2 , and q refers to the order of the ARCH terms ϵ^2 . The conditional variance of the GARCH(1,1) used in this study is given by:

$$\sigma_t^2 = \omega + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2 \quad (6)$$

Nelson (1991) extended the standard GARCH framework by introducing the EGARCH, with the conditional variance of the EGARCH (1,1) expressed as:

$$\ln \sigma_t^2 = \omega + \beta_1 \ln \sigma_{t-1}^2 + \alpha_1 \left| \frac{\epsilon_{t-1}}{\sigma_{t-1}} \right| + \gamma \frac{\epsilon_{t-1}}{\sigma_{t-1}} \quad (9)$$

Table 6. GARCH estimation output.

Selected GARCH model	RKLCI AR(1)-EGARCH (1,1)	RLQ ARMA(1,3)-EGARCH(1,1)	RSET AR(2) AR(6) MA(1)-EGARCH(1,1)
Mean equation			
C	0.0000	0.0002	0.0003
AR	0.107	0.025 -0.069	0.031 -0.043 0.068
Variance equation			
Constant, ω	-0.332	-0.273	-0.333
ARCH term, α	0.165	0.149	0.203
Leverage effect term, γ	-0.080	-0.076	-0.072
GARCH term, β	0.980	0.982	0.981
Log-likelihood	10831.380	8585.716	9499.327
Heteroscedasticity	No	No	No
Autocorrelation	No	No	No

Methods – T-GARCH: Lim et al. (2023)

The TGARCH(1,1), as proposed by [Zakoian \(1994\)](#), has the following conditional variance equation:

$$\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \gamma d_{t-1} \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 \quad (10)$$

Where negative ε_t is represented by a binary decision variable:

$$d_{t-1} = \begin{cases} 1 & \text{if } \varepsilon_{t-1} < 0 \text{ (negative shocks)} \\ 0 & \text{if } \varepsilon_{t-1} \geq 0 \text{ (positive shocks)} \end{cases}$$

In the case where $\gamma = 0$, the TGARCH(1,1) model is reduced to the GARCH(1,1) model.

Methods – DCC-GARCH: Le and Tran (2021)

The Contagion Effect from U.S. Stock Market to the Vietnamese and the Philippine Stock Markets: The Evidence of DCC – GARCH Model

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Received: November 05, 2020 Revised: December 30, 2020 Accepted: January 15, 2021

Abstract

Using a DCC – GARCH model analysis, this paper examines the existence of financial contagion from the U.S. stock market to the Vietnamese and the Philippine stock markets during the global financial crisis and the COVID-19 pandemic crisis. We use daily data from the S&P 500 (U.S.), VN-Index (Vietnam), and the PSEi (the Philippines). As a result, there is no evidence of contagion from the U.S stock market to the Philippine stock market that can be found during global financial crisis, while the Vietnamese market is influenced by this effect. Besides, both these developing stock markets (the Vietnamese and Philippine stock markets) are influenced by the contagion effect in COVID-19 pandemic crisis. Another finding is that the contagion effect during the coronavirus pandemic crisis in Vietnam is smaller than that during the global financial crisis, however, the opposite is the case for the Philippines. It is noticed that the Philippines seems to be more affected by the contagion effect from the COVID-19 pandemic than Vietnam at the time of this study. Because financial contagion is important for monetary policy, asset pricing, risk measurement, and portfolio allocation, the findings in this paper may give some useful information for policymakers and investors.

Keywords: Contagion, COVID-19, Financial Crisis, DCC – GARCH, Stock Market

Methods – DCC-GARCH: Le and Tran (2021).

The multivariate DCC–GARCH model that is proposed by Engle (2002) is defined as follows;

$$Y_t = \mu_t + H_t^{1/2} \varepsilon_t \quad (1)$$

$$\begin{cases} H_t = D_t R_t D_t \\ R_t = (\text{diag}(Q_t))^{-1/2} Q_t (\text{diag}(Q_t))^{-1/2} \\ D_t = \text{diag}(\sqrt{h_{11,t}}, \sqrt{h_{22,t}}, \dots, \sqrt{h_{NN,t}}) \end{cases} \quad (2)$$

where $Y_t = (Y_{1t}, Y_{2t}, Y_{3t}, \dots, Y_{Nt})$ is the vector of the past observations, $\mu_t = (\mu_{1t}, \mu_{2t}, \mu_{3t}, \dots, \mu_{Nt})$ is the vector of conditional returns, H_t is the multivariate conditional variance, $\varepsilon_t = (\varepsilon_{1t}, \varepsilon_{2t}, \varepsilon_{3t}, \dots, \varepsilon_{Nt})$ is the vector of the standardized residuals, R_t is a $N \times N$ symmetric dynamic correlations matrix, and D_t is a diagonal matrix of conditional standard deviations for return series, which is got from analyzing a GARCH model with $\sqrt{h_{ii,t}}$ on the diagonal $i(i=1, 2, 3, \dots, N)$.

The DCC specification is defined as follows;

$$\begin{aligned} Q_t &= (1 - \lambda_1 - \lambda_2) \bar{Q} + \lambda_1 \delta_{i,t-1} \delta_{j,t-1} + \lambda_2 Q_{t-1}^* \\ R_t &= Q_t^{-1} Q_t Q_t^{-1} \end{aligned} \quad (3)$$

where $Q_t = [q_{ij,t}]$ is a $(N \times N)$ time varying covariance matrix of standardized residuals $\left(\delta_{it} = \frac{\varepsilon_{it}}{\sqrt{h_{it}}} \right)$, $\bar{Q}$ is the unconditional correlations of $\delta_{i,t} \delta_{j,t}$, λ_1 and λ_2 are nonnegative scalar parameters which satisfies $\lambda_1 + \lambda_2 < 1$, and $Q_t^* = [q_{ii,t}^*] = \sqrt{q_{ii,t}}$ is a diagonal matrix with the square root of the i^{th} diagonal element of Q_t on its i^{th} diagonal position.

Hence, the conditional correlation of a pair of markets i and j at time t can be defined as:

$$\rho_{ij,t} = \frac{(1 - \lambda_1 - \lambda_2) \bar{q}_{ij} + \lambda_1 \delta_{i,t-1} \delta_{j,t-1} + \lambda_2 q_{ij,t-1}}{\left[\frac{[(1 - \lambda_1 - \lambda_2) \bar{q}_{ii} + \lambda_1 \delta_{i,t-1}^2 + \lambda_2 q_{ii,t-1}]^{\frac{1}{2}} \times [(1 - \lambda_1 - \lambda_2) \bar{q}_{jj} + \lambda_1 \delta_{j,t-1}^2 + \lambda_2 q_{jj,t-1}]^{\frac{1}{2}}}{2} \right]} \quad (4)$$

where q_{ij} is the element which locates on the i^{th} and j^{th} column in the matrix Q_t .

Methods – DCC-GARCH: Le and Tran (2021)

Table 3: Analysis of DCC – GARCH model

		Dynamic conditional correlation		
		Pre-crisis	Crisis	% difference
Global Financial crisis	US_Vietnam	0.0596* (0.077)	0.1143** (0.012)	91.7785
	US_Philippines	0.0631* (0.097)	0.0415 (0.399)	-34.2314
COVID-19 pandemic crisis	US_Vietnam	0.3431*** (0.001)	0.4199*** (0.000)	22.3841
	US_Philippines	0.2128** (0.036)	0.3247*** (0.000)	52.5846

Note: pre-crisis period is from 03/01/2005 to 31/09/2007 and from 03/09/2019 to 23/02/2020. Crisis period is from 01/10/2007 to 31/12/2009 and from 24/02/2020 to 30/06/2020. ***, ** and * indicate the significance level at 1%, 5%, 10% respectively.

5.2. The Contagion Effects

Table 3 presents the analysis results of DCC coefficients in the period of pre-crisis and the period of crisis in global financial crisis and COVID-19 pandemic crisis. According to the statistics during the 2007–2008 financial crisis, the DCC coefficients between the U.S. and Vietnamese markets in the pre-crisis period and crisis period are significant at 10% level and 5% level, respectively, however, that value from the U.S. to the Philippine markets is only significant at 10% level in the pre-crisis period, and is not significant at all the levels in the crisis period, implying that the correlation between two markets is very weak, and almost none. It can be clearly seen that DCC correlation between the U.S. and the Vietnamese stock markets increases in crisis period, while this correlation between the U.S. and the rest of market falls, which also shows the difference based on the

Methods – DCC-GARCH: Yadav et al. (2023)

Research

Volatility Spillover Effects between Indian Stock Market and Global Stock Markets: A DCC-GARCH Model

FIB Business Review

1–13

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Nikhil Yadav¹, Anurag Bhadur Singh²  and Priyanka Tandon³ 

Abstract

The present article empirically estimates the volatility spillover transmission in Indian equity market represented by Sensex from world economies composite index (Euro Stoxx 50) using the dynamic conditional correlation generalized autoregressive conditional heteroscedasticity (DCC-GARCH) model. The study uses secondary data spanning between 1 April 2012 and March 2022 on weekly basis. The DCC-GARCH model is applied to examine the spillover from developed stock markets to Indian stock market (Sensex). The findings of the study revealed that in short run there is a spillover effect from global markets to Indian stock markets. Investors can invest in the Indian stock market for the long period of time as there is no volatility spillover or volatility transmission from Euro and Nasdaq however in short run the investment in the Indian stock market is not safe due to the presence of volatility effect from all developed stock markets.

Keywords

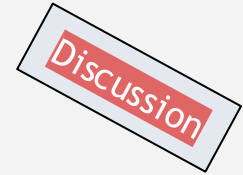
Volatility spillovers, return, price discovery, ARCH model, DCC-GARCH model and Sensex

$$R_{it} = a + b_1 R_{t-1} + \varepsilon_{it} = h_{it}^{1/2} v_{it} \quad (3)$$

$$C_t = D_t R_t D_t \quad (4)$$

$$p_{y,t} = \frac{q_{y,t}}{\sqrt{q_{ii,t} q_{jj,t}}} \quad (8)$$

Methods – DCC-GARCH: Yadav et al. (2023)



How to interpret the following estimation results?

Table 5. Results of DCC-GARCH Model.

Panel (5A): DCC from reuro to rsensex

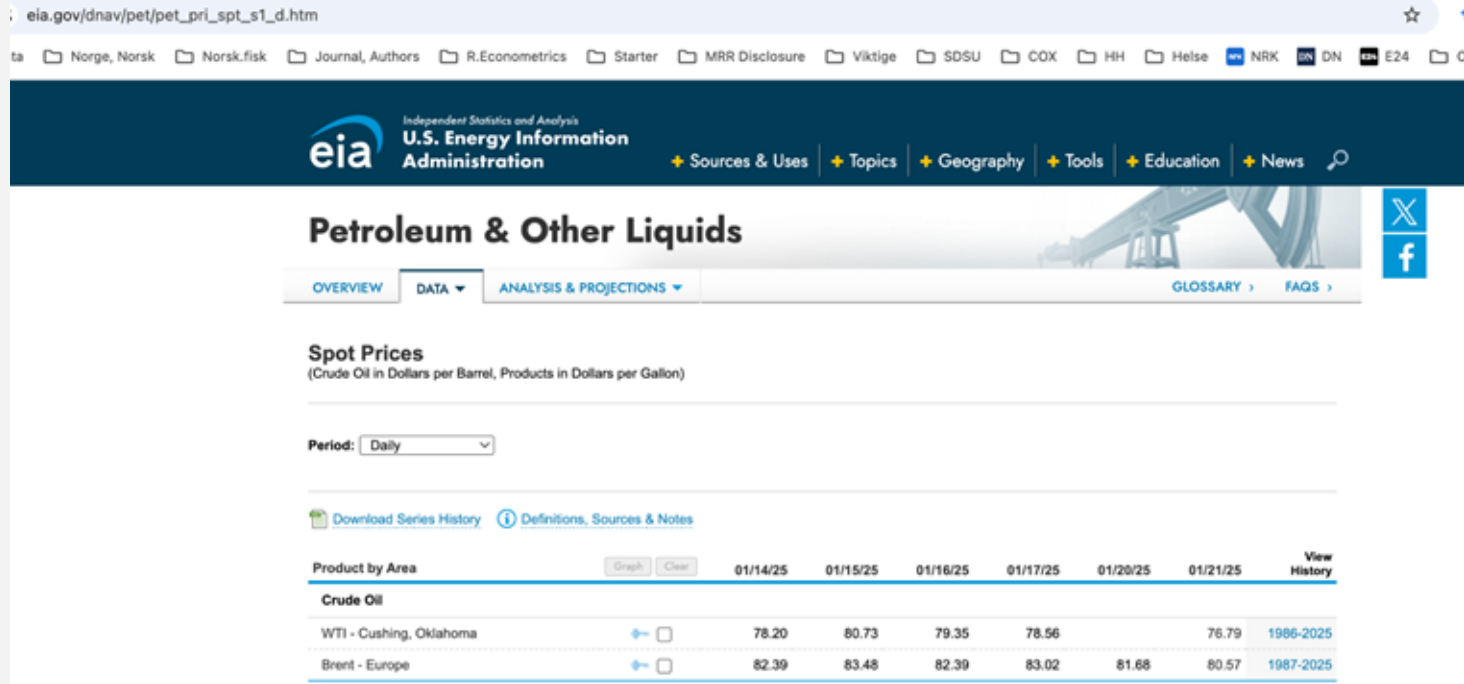
	Estimate	Std. Error	t-Value	Pr(> t)
[nasdaq].mu	4.4324	3.1642	1.4008	0.1612
[nasdaq].omega	565.3375	310.1200	1.8229	0.0683***
[nasdaq].alpha1	0.1604	0.0801	2.0006	0.0454**
[nasdaq].beta1	0.7738	0.0889	8.6961	0.0000*
[rsensex].mu	75.2063	30.9650	2.4287	0.0151**
[rsensex].omega	597.1252	2371.4	0.2518	0.8011
[rsensex].alpha1	0.0906	0.0261	3.4651	0.0005*
[rsensex].beta1	0.9083	0.0330	27.4476	0.0000*
[joint]dcca1	0.0000	0.0001	0.0000	0.9999
[joint]dccb1	0.9170	0.2648	3.4623	0.0005*

to Sensex, panel (5B) presents the spillover effect from Nasdaq to Sensex and panel (5C) presents the spillover effect from NYSE to Sensex. The DCC-alpha (DCCa1) signifies the spillover effect in the short run, whereas the DCC-beta (DCCb1) signifies the spillover effect in long run. The dcca1 value is insignificant across all panels in the study, which clearly indicates that there is a presence of no volatility spillover or volatility transmission from global financial markets to the Indian stock market proxied by Sensex in short-run. However, dccb1 parameter for Panel (5A) is insignificant (0.0000), which indicates that there is volatility transmission from reuro to Sensex, that is, Eurozone financial markets to Indian stock market in long-run period. The findings of the study postulate that investor can invest in the Indian stock market for short-run period as there is no sign of volatility spillover from the global stock market considered in the study. But, at the same time,

Do you agree with the authors' interpretation?

Part 4: Data and key variables

Spot prices of crude oil



Oil companies

☆ Sedigh et al. (2019).... x

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oil price, stock price and stock returns have been collected from www.nyse.com, www.nasdaq.com and www.marketwatch.com.

Table 1. Petroleum Companies.

Name	Stock Symbol	Name	Stock Symbol
ExxonMobil	NYSE: XOM	Southwestern Energy	NYSE: SWN
Vaalco Energy	NYSE: EGY	Hess Corporation	NYSE: HES
Whiting Petroleum	NYSE: WLL	Range Resources	NYSE: RRC
WPX Energy	NYSE: WPX	Linn Energy	NASDAQ: LNGG
Rex Energy	NASDAQ: REXX	Pioneer Natural Resources	NYSE: PXD
Noble Energy	NYSE: NBL	Murphy Oil	NYSE: MUR
Laredo Petroleum	NYSE: LPI	Newfield Exploration	NYSE: NFX
Occidental Petroleum	NYSE: OXY	Concho Resources	NYSE: CXO
ConocoPhillips	NYSE: COP	Oasis Petroleum	NYSE: OAS
Continental Resources	NYSE: CLR	Marathon Oil	NYSE: MRO
Denbury Resources	NYSE: DNR	SandRidge Energy	NYSE: SD
Devon Energy	NYSE: DVN	SM Energy	NYSE: SM
Energen	NYSE: EGN	Chesapeake Energy	NYSE: CHK
EOG Resources	NYSE: EOG	Chevron Corporation	NYSE: CVX
EQT	NYSE: EQT	Cimarex Energy	NYSE: XEC

NYSE - Nasdaq Real Time Price - USD

Equinor ASA (EQNR) ☆

35.25 +0.88 (+2.56%) 35.40 +0.15 (+0.43%)

At close: March 13 at 4:00:02 PM EDT

After hours: March 13 at 7:55:15 PM EDT

Oil companies

Company Name	Ticker	Yahoo Finance Link
Exxon Mobil Corporation	XOM	https://finance.yahoo.com/quote/XOM
Chevron Corporation	CVX	https://finance.yahoo.com/quote/CVX
ConocoPhillips	COP	https://finance.yahoo.com/quote/COP
BP p.l.c.	BP	https://finance.yahoo.com/quote/BP
Shell plc	SHEL	https://finance.yahoo.com/quote/SHEL
TotalEnergies SE	TTE	https://finance.yahoo.com/quote/TTE
Marathon Oil Corporation	MRO	https://finance.yahoo.com/quote/MRO
Valero Energy Corporation	VLO	https://finance.yahoo.com/quote/VLO
Phillips 66	PSX	https://finance.yahoo.com/quote/PSX
Occidental Petroleum Corporation	OXY	https://finance.yahoo.com/quote/OXY

New energy company

- **NextEra Energy (NYSE: NEE):** Headquartered in Juno Beach, Florida, NextEra Energy is the world's largest generator of renewable energy from wind and solar. Through its subsidiary, NextEra Energy Resources, it operates numerous renewable energy facilities across North America.

NextEra Energy, Inc.	NEE	https://finance.yahoo.com/quote/NEE
Brookfield Renewable Partners L.P.	BEP	https://finance.yahoo.com/quote/BEP
Clearway Energy, Inc.	CWEN	https://finance.yahoo.com/quote/CWEN
First Solar, Inc.	FSLR	https://finance.yahoo.com/quote/FSLR
SolarEdge Technologies, Inc.	SEDG	https://finance.yahoo.com/quote/SEDG
Sunrun Inc.	RUN	https://finance.yahoo.com/quote/RUN
Sunnova Energy International Inc.	NOVA	https://finance.yahoo.com/quote/NOVA

Raw Data

R-Code

```
library(quantmod) # ext
library(readxl)   # Imp
library(tseries)
library(FinTS)    # for AR
library(rugarch)  # ARCH
library(rmgarch)  # DCC
```

```
## Read and merge data
#####

D.Oil <- read_excel("PET_PRI_SPT_S1_D.xlsx", sheet='tilR')
D.Oil = as.data.frame(D.Oil); nrow(D.Oil) # head(D.Oil)
D.Oil = D.Oil[complete.cases(D.Oil),]; nrow(D.Oil)

getSymbols("XOM", src = "yahoo", from = "2010-01-01", to = "2024-01-01")
XOM1 = data.frame(row.names(data.frame(XOM)), XOM[,6]); head(XOM1)
names(XOM1) = c('Date', 'XOM'); row.names(XOM1) = NULL
XOM1 = XOM1[complete.cases(XOM1),]; nrow(XOM1)

getSymbols("NEE", src = "yahoo", from = "2010-01-01", to = "2024-01-01")
# head(NEE); nrow(NEE)
NEE1 = data.frame(row.names(data.frame(NEE)), NEE[,6]); head(NEE1)
names(NEE1) = c('Date', 'NEE'); row.names(NEE1) = NULL
NEE1 = NEE1[complete.cases(NEE1),]; nrow(NEE1)

nrow(XOM1); nrow(NEE1); nrow(D.Oil)

Da = merge(XOM1,NEE1,by='Date' ); nrow(Da); head(Da)
D.Oil$Date = as.character(D.Oil$Date)

Da2 = merge(Da,D.Oil,by='Date' ); nrow(Da2)
Da2 = Da2[Da2$Oil > 0, ]
Da2 = Da2[order(Da2$Date), ]

summary(Da2)
```

```
X0 = diff(log(Da2$Oil))*100;
Y0 = diff(log(Da2$XOM))*100;
Z0 = diff(log(Da2$NEE))*100;
Date = as.Date(Da2$Date[2:nrow(Da2)])
```

```
Da3 = data.frame(Date,X0, Y0, Z0);
summary(Da3)
```

```
## delete outliers
sort(Da3$X)[1:50]
sort(Da3$X, decreasing = TRUE)[1:50]
```

```
Da4 = Da3[Da3$X0 < 10 & Da3$X0 > -10,]
Da4 = Da4[Da4$Y0 < 10 & Da4$Y0 > -10,]
Da4 = Da4[Da4$Z0 < 10 & Da4$Z0 > -10,]
```

```
summary(Da4)
```

```
# Convert variables into a Time Series Object
## Cannot use TS for the reason of missing values
#####
```

```
X <- zoo(Da4$X0, order.by = Da4$Date)
X <- as.xts(X)
```

```
Y <- zoo(Da4$Y0, order.by = Da4$Date)
Y <- as.xts(Y)
```

```
Z <- zoo(Da4$Z0, order.by = Da4$Date)
Z <- as.xts(Z)
```

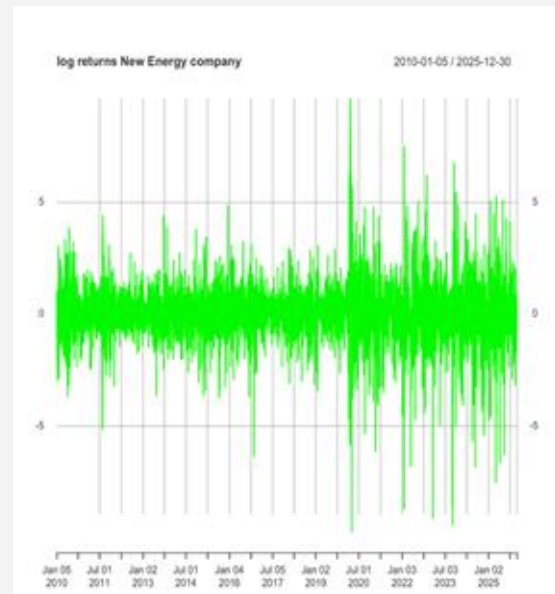
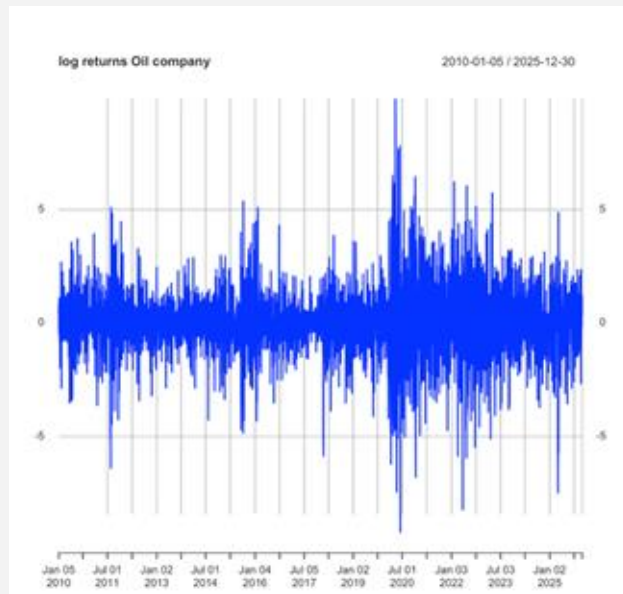
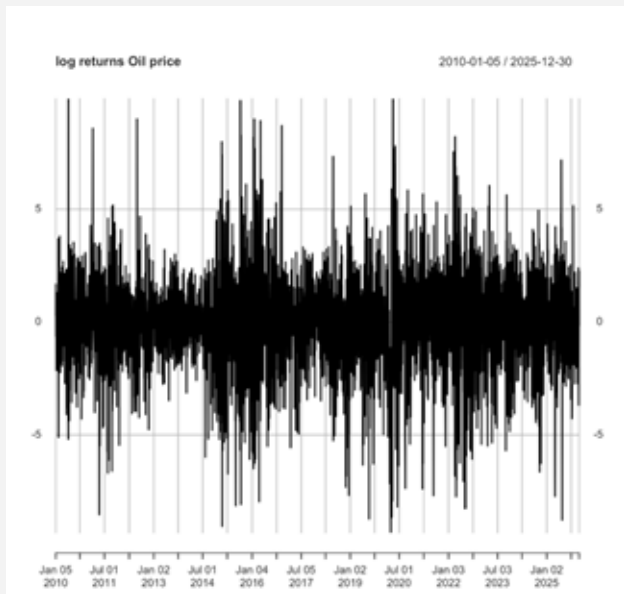
Raw Data

```
> summary(Da2)
```

Date	XOM	NEE	Oil
Length:4008	Min. : 23.99	Min. : 7.095	Min. : 8.91
Class :character	1st Qu.: 49.16	1st Qu.:15.661	1st Qu.: 54.18
Mode :character	Median : 54.73	Median :31.351	Median : 71.75
	Mean : 62.24	Mean :38.352	Mean : 71.69
	3rd Qu.: 62.74	3rd Qu.:65.089	3rd Qu.: 88.30
	Max. :120.19	Max. :85.706	Max. :123.64

```
> |
```

Log returns



Part 5: Estimation and interpretation – Examples

“For your replication”

Sample companies

- Oil price (X)
- XOM (Y)
- NEE (Z)

```
> summary(Da4)
```

Date	X0	Y0	Z0
Min. :2010-01-05	Min. : -9.337407	Min. : -9.23925	Min. : -9.76693
1st Qu.:2013-12-16	1st Qu.: -1.164397	1st Qu.: -0.71490	1st Qu.: -0.60682
Median :2017-12-02	Median : 0.079500	Median : 0.02250	Median : 0.11001
Mean :2017-12-20	Mean : -0.002071	Mean : 0.03551	Mean : 0.06491
3rd Qu.:2022-01-02	3rd Qu.: 1.212730	3rd Qu.: 0.80528	3rd Qu.: 0.76168
Max. :2025-12-30	Max. : 9.903557	Max. : 9.89033	Max. : 9.64517

```
> var(Y);  
      [,1]  
[1,] 2.188512  
> var(Z)  
      [,1]  
[1,] 1.80812  
> cor(Y,X);  
      [,1]  
[1,] 0.4956046  
> cor(Z,X)  
      [,1]  
[1,] 0.04445315
```

Results: ARCH Test

```
> arch_test <- ArchTest(X, lags = 5); print(arch_test)

      ARCH LM-test; Null hypothesis: no ARCH effects

data:  X
Chi-squared = 396.54, df = 5, p-value < 2.2e-16

> arch_test <- ArchTest(Y, lags = 5); print(arch_test)

      ARCH LM-test; Null hypothesis: no ARCH effects

data:  Y
Chi-squared = 539.2, df = 5, p-value < 2.2e-16

> arch_test <- ArchTest(Z, lags = 5); print(arch_test)

      ARCH LM-test; Null hypothesis: no ARCH effects

data:  Z
Chi-squared = 397.02, df = 5, p-value < 2.2e-16

> |
```

Results: ARCH for Y

- Omega (ω): The constant term in the variance equation.
- Alpha (α): Measures the impact of past shocks (ARCH effect).

```
*-----*
*          GARCH Model Fit          *
*-----*
```

Conditional Variance Dynamics

```
-----
GARCH Model      : sGARCH(1,0)
Mean Model       : ARFIMA(0,0,0)
Distribution      : norm
```

Optimal Parameters

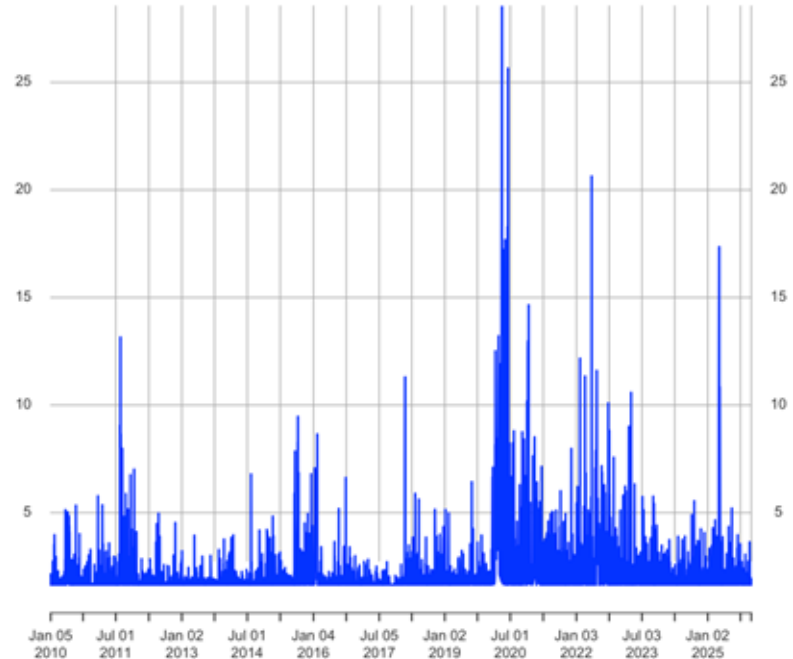
```
-----
      Estimate  Std. Error  t value Pr(>|t|)
mu      0.053366    0.021289   2.5067 0.012185
omega    1.601589    0.048132  33.2748 0.000000
alpha1   0.278499    0.027291  10.2048 0.000000
```

Robust Standard Errors:

```
      Estimate  Std. Error  t value Pr(>|t|)
mu      0.053366    0.02157   2.4741 0.013358
omega    1.601589    0.11358  14.1013 0.000000
alpha1   0.278499    0.05303   5.2518 0.000000
```

Conditional Variance (ARCH model)

2010-01-05 / 2025-12-30



Results: ARCH for Z

- Omega (ω): The constant term in the variance equation.
- Alpha (α): Measures the impact of past shocks (ARCH effect).

```
*-----*
*           GARCH Model Fit           *
*-----*
```

Conditional Variance Dynamics

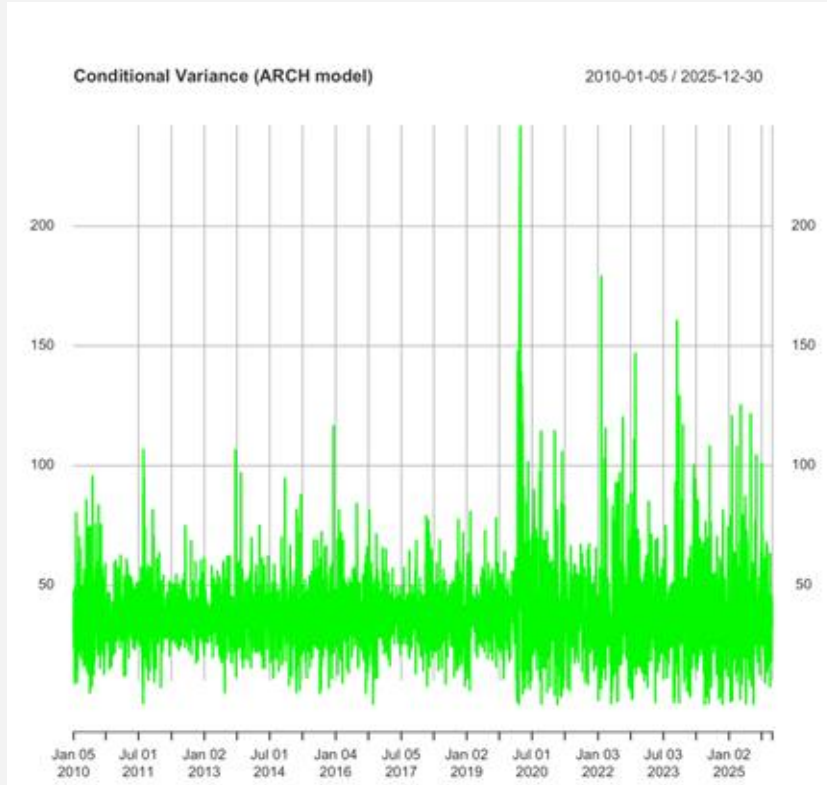
```
-----
GARCH Model      : sGARCH(1,0)
Mean Model       : ARFIMA(0,0,0)
Distribution      : norm
```

Optimal Parameters

```
-----
      Estimate  Std. Error  t value  Pr(>|t|)
mu      -5.944933    0.003882   -1531.3    0
omega    0.028743    0.000013    2200.9    0
alpha1   0.995249    0.000453    2197.2    0
```

Robust Standard Errors:

```
      Estimate  Std. Error  t value  Pr(>|t|)
mu      -5.944933    435.5592   -0.013649  0.98911
omega    0.028743     1.5659    0.018356  0.98535
alpha1   0.995249     54.2651    0.018340  0.98537
```



Results: GARCH for Y

1. Univariate GARCH Parameters (for each asset in the system)

- **Omega (ω):** The constant term in the variance equation.
- **Alpha (α):** Measures the impact of past shocks (ARCH effect).
- **Beta (β):** Measures the persistence of volatility (GARCH effect).

```
> print(garch_fit)
```

```
*-----*
*      GARCH Model Fit      *
*-----*
```

Conditional Variance Dynamics

```
-----
GARCH Model      : sGARCH(1,1)
Mean Model       : ARFIMA(0,0,0)
Distribution      : norm
```

Optimal Parameters

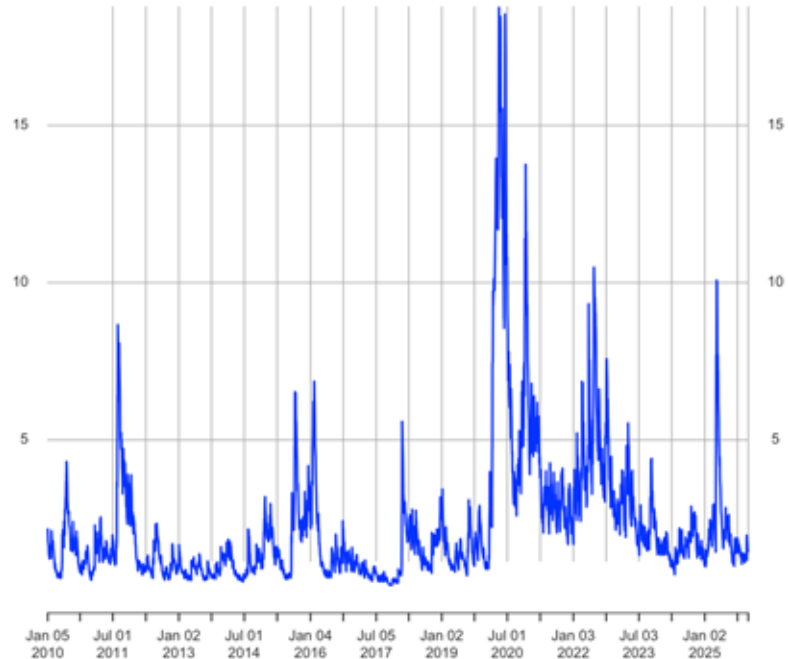
```
-----
      Estimate Std. Error t value Pr(>|t|)
mu      0.035367   0.017842   1.9822 0.047459
omega    0.020745   0.004900   4.2337 0.000023
alpha1   0.077362   0.008460   9.1443 0.000000
beta1    0.914479   0.008995 101.6620 0.000000
```

Robust Standard Errors:

```
      Estimate Std. Error t value Pr(>|t|)
mu      0.035367   0.017950   1.9702 0.048810
omega    0.020745   0.007583   2.7359 0.006221
alpha1   0.077362   0.013575   5.6989 0.000000
beta1    0.914479   0.014508  63.0308 0.000000
```

Conditional Variance (GARCH model)

2010-01-05 / 2025-12-30



Results: GARCH for Z

1. Univariate GARCH Parameters (for each asset in the system)

- **Omega (ω):** The constant term in the variance equation.
- **Alpha (α):** Measures the impact of past shocks (ARCH effect).
- **Beta (β):** Measures the persistence of volatility (GARCH effect).

```
*-----*
*           GARCH Model Fit           *
*-----*
```

Conditional Variance Dynamics

```
-----
GARCH Model      : sGARCH(1,1)
Mean Model       : ARFIMA(0,0,0)
Distribution      : norm
```

Optimal Parameters

```
-----
```

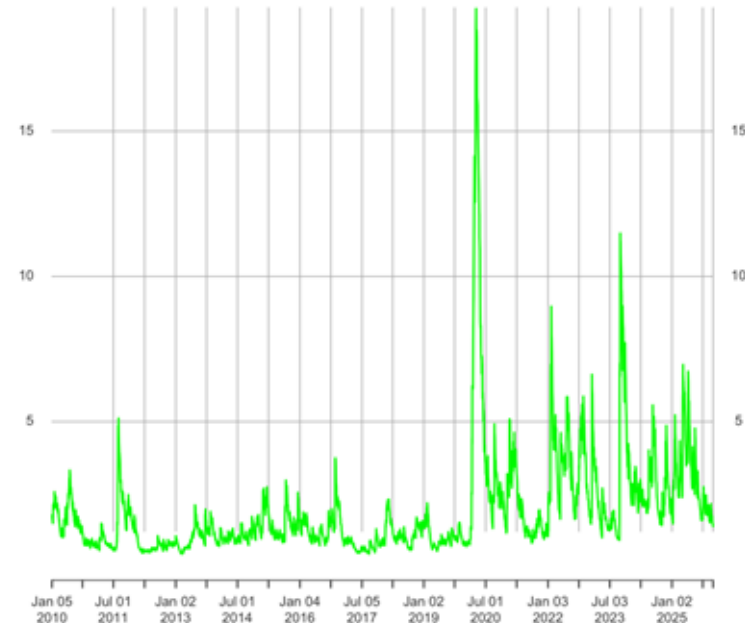
	Estimate	Std. Error	t value	Pr(> t)
mu	0.083771	0.017120	4.8933	0.000001
omega	0.014432	0.004450	3.2432	0.001182
alpha1	0.059229	0.007973	7.4283	0.000000
beta1	0.934900	0.008904	105.0036	0.000000

Robust Standard Errors:

	Estimate	Std. Error	t value	Pr(> t)
mu	0.083771	0.016389	5.1114	0.000000
omega	0.014432	0.012037	1.1990	0.230540
alpha1	0.059229	0.028163	2.1031	0.035457
beta1	0.934900	0.031622	29.5646	0.000000

Conditional Variance (GARCH model)

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Results: T-GARCH for Y

$$h_t = \delta + \alpha_1 e_{t-1}^2 + \gamma d_{t-1} e_{t-1}^2 + \beta_1 h_{t-1}$$

$$d_t = \begin{cases} 1 & e_t < 0 \text{ (bad news)} \\ 0 & e_t \geq 0 \text{ (good news)} \end{cases}$$

```

*-----*
*           GARCH Model Fit           *
*-----*

Conditional Variance Dynamics
-----
GARCH Model      : fgARCH(1,1)
FGARCH Sub-Model : TGARCH
Mean Model       : ARFIMA(0,0,0)
Distribution      : norm

Optimal Parameters
-----

```

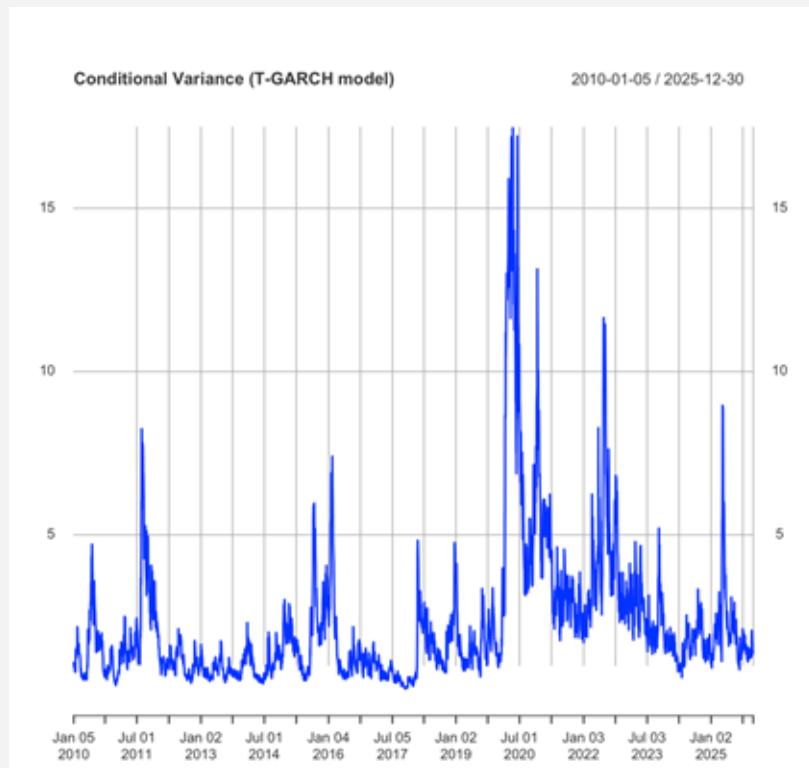
	Estimate	Std. Error	t value	Pr(> t)
mu	0.021395	0.016885	1.2671	0.205110
omega	0.018964	0.004202	4.5132	0.000006
alpha1	0.089931	0.009066	9.9199	0.000000
beta1	0.917526	0.008756	104.7857	0.000000
eta11	0.255570	0.052800	4.8403	0.000001

```

Robust Standard Errors:

```

	Estimate	Std. Error	t value	Pr(> t)
mu	0.021395	0.016382	1.3060	0.191559
omega	0.018964	0.007136	2.6577	0.007868
alpha1	0.089931	0.015767	5.7037	0.000000
beta1	0.917526	0.015949	57.5304	0.000000
eta11	0.255570	0.067957	3.7608	0.000169



Results: T-GARCH for Z

* GARCH Model Fit *

Conditional Variance Dynamics

GARCH Model : fGARCH(1,1)
fGARCH Sub-Model : TGARCH
Mean Model : ARFIMA(0,0,0)
Distribution : norm

Optimal Parameters

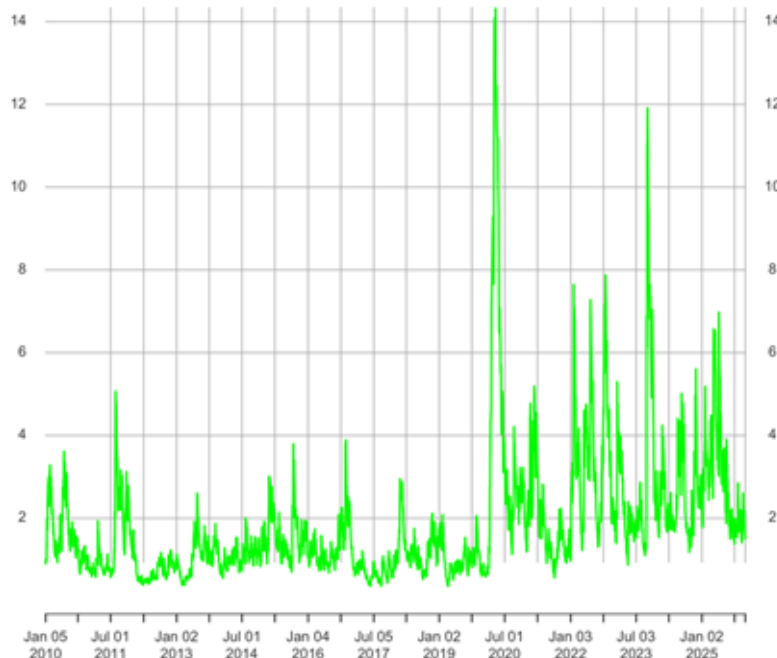
	Estimate	Std. Error	t value	Pr(> t)
mu	0.064155	0.019339	3.3174	0.000908
omega	0.014597	0.004752	3.0718	0.002128
alpha1	0.076329	0.009667	7.8957	0.000000
beta1	0.932330	0.009935	93.8425	0.000000
eta11	0.382516	0.068136	5.6140	0.000000

Robust Standard Errors:

	Estimate	Std. Error	t value	Pr(> t)
mu	0.064155	0.022457	2.85682	0.004279
omega	0.014597	0.014612	0.99899	0.317802
alpha1	0.076329	0.038538	1.98063	0.047633
beta1	0.932330	0.039012	23.89842	0.000000
eta11	0.382516	0.140656	2.71952	0.006538

Conditional Variance (T-GARCH model)

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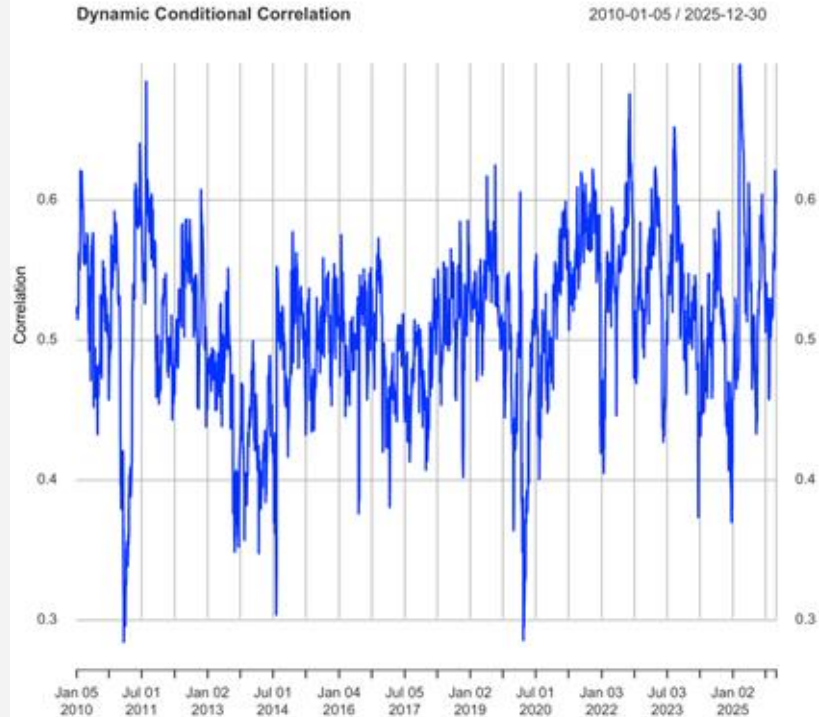


$$Q_t = (1 - a - b)\bar{Q} + a(z_{t-1}z_{t-1}^\top) + bQ_{t-1}$$

Results: DCC-GARCH (x-y)

Optimal Parameters

	Estimate	Std. Error	t value	Pr(> t)
[X].mu	0.024408	0.029693	0.82202	0.411065
[X].omega	0.097853	0.030947	3.16192	0.001567
[X].alpha1	0.084339	0.013262	6.35933	0.000000
[X].beta1	0.895935	0.017218	52.03614	0.000000
[Y].mu	0.035367	0.018584	1.90308	0.057029
[Y].omega	0.020745	0.007524	2.75726	0.005829
[Y].alpha1	0.077362	0.013509	5.72664	0.000000
[Y].beta1	0.914479	0.014886	61.43202	0.000000
[Joint]dcca1	0.019555	0.006409	3.05118	0.002279
[Joint]dccb1	0.947857	0.016682	56.81855	0.000000

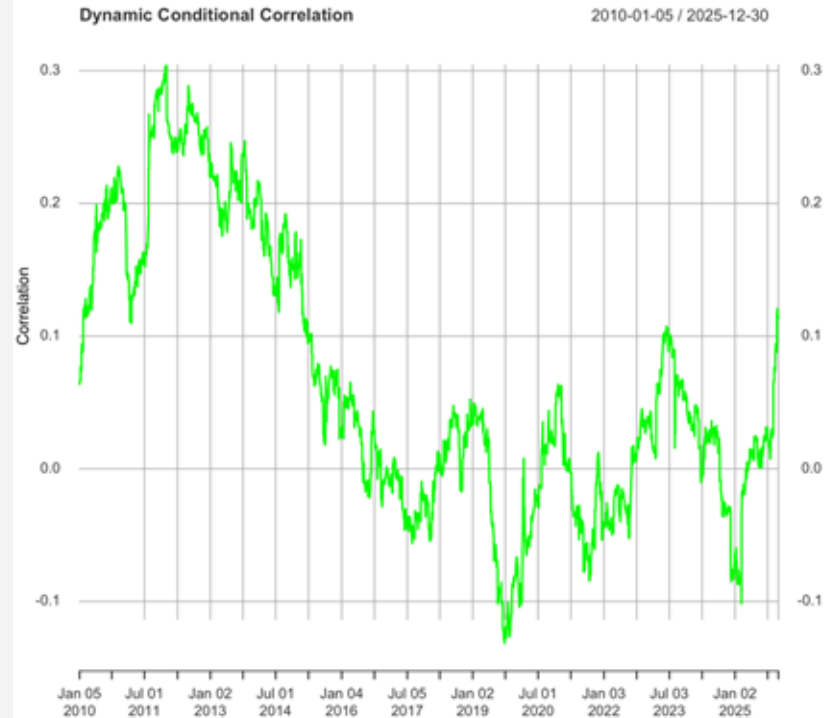


$$Q_t = (1 - a - b)\bar{Q} + a(z_{t-1}z_{t-1}^\top) + bQ_{t-1}$$

Results: DCC-GARCH (x-z)

Optimal Parameters

	Estimate	Std. Error	t value	Pr(> t)
[X].mu	0.024408	0.029683	0.82228	0.410916
[X].omega	0.097853	0.031089	3.14751	0.001647
[X].alpha1	0.084339	0.013276	6.35283	0.000000
[X].beta1	0.895935	0.017292	51.81318	0.000000
[Z].mu	0.083771	0.017746	4.72070	0.000002
[Z].omega	0.014432	0.010632	1.35742	0.174647
[Z].alpha1	0.059229	0.022221	2.66542	0.007689
[Z].beta1	0.934900	0.026151	35.75008	0.000000
[Joint]dcca1	0.004447	0.001879	2.36622	0.017971
[Joint]dccb1	0.994940	0.002183	455.67422	0.000000



Comparison for Y

Optimal Parameters

	Estimate
mu	0.053366
omega	1.601589
alpha1	0.278499

Optimal Parameters

	Estimate
mu	0.035367
omega	0.020745
alpha1	0.077362
beta1	0.914479

Optimal Parameters

	Estimate
mu	0.021395
omega	0.018964
alpha1	0.089931
beta1	0.917526
eta11	0.255570

Optimal Parameters

	Estimate
[X].mu	0.024408
[X].omega	0.097853
[X].alpha1	0.084339
[X].beta1	0.895935
[Y].mu	0.035367
[Y].omega	0.020745
[Y].alpha1	0.077362
[Y].beta1	0.914479
[Joint]dccca1	0.019555
[Joint]dcccb1	0.947857

Reversion to the Steady-State in a DCC-GARCH

- **High α , Low β** $\rightarrow$ Correlation reacts quickly to new information but does not persist for long.
- **Low α , High β** $\rightarrow$ Correlation moves slowly in response to shocks and takes longer to revert to steady-state.
- **High $\alpha+\beta$ (close to 1)** $\rightarrow$ Strong persistence, meaning correlation reverts **slowly**.
- **Low $\alpha+\beta$** $\rightarrow$ Correlation reverts **quickly**.

Half-Life

2. Half-Life for the DCC-GARCH Model

For the DCC-GARCH model:

$$Q_t = (1 - \alpha - \beta)Q^* + \alpha(\epsilon_{t-1}\epsilon_{t-1}^T) + \beta Q_{t-1}$$

The half-life of correlation reversion is given by:

$$t_{1/2} = \frac{\ln(0.5)}{\ln(\alpha + \beta)}$$

where $\alpha + \beta$ determines the persistence of conditional correlations.

- If $\alpha + \beta$ is **close to 1**, the half-life is **long** (slow decay).
- If $\alpha + \beta$ is **small**, the half-life is **short** (fast decay).

Part 6: Preliminary results and presentation

Presentation

- Sample companies
 - Introduce the oil company and/or new energy company you chose.
- Choose one of the hypotheses:
 - Hypothesis 1 (H1): There are significant ARCH and GARCH effects for stock returns of the traditional oil company and the renewable energy company.
 - Hypothesis 2 (H2): The conditional correlation between oil price changes and stock returns of the traditional oil companies exhibits both short-term and long-term persistence.
 - Hypothesis 3 (H3): The conditional correlation between oil price changes and stock returns of the new energy company exhibits both short-term and long-term persistence.
- Present the methods:
 - Univariate: ARCH, GARCH, T-GARCH
 - Multivariate: DCC-GARCH
- Primary results:
 - Summary Statistics of stock returns of the two companies
 - Test ARCH effects for the the two stock prices
 - Univariate: ARCH, GARCH, T-GARCH
 - Multivariate: DCC-GARCH

Presentation

- 10 minutes presentation
- 5 minutes for comments and suggestions.